



CHARUSAT[®]
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

STUDENT INFORMATION BOOKLET

VOLUME – 2:

Bachelor of Science in Information Technology

Smt. Chandaben Mohanbhai Patel Institute of
Computer Applications

CHARUSAT UNIVERSITY

Off. Nadiad-Petlad Highway, Changa - 388 421
Anand, Gujarat, India

CONTENT

Sr. No.	Description	Pg. No.
	Preamble	1
	About the Institute	2
1	PROGRAMME SPECIFIC	
1.1	Academic Regulations	3
1.2	Programme Structure (Teaching & Examination Scheme)	21
1.3	Academic Calendar	152
2	VARIOUS ADMINISTRATIVE PROCESS	
2.1	Payment of tuition fees or Other charges	155
2.2	Process of Acquiring WIFI access	156
2.3	Process to obtain required Certificate from the institute	157
2.4	Process to obtain required Certificate from the university office	158
2.5	Process to obtain Duplicate Grade Card / Name Correction in Grade Card	159
2.6	Process to obtain Migration Certificate	159
2.7	Process for acquiring Transcript OR E-transcript	159
2.8	Process for Academic Document Verification by External Agency	160
3	FORMS FOR UNDERTAKINGS & DECLARATIONS	
3.1	Undertaking for Observing Rules and Regulations of the university	162
3.2	Declaration for Code of Conducts & Disciplinary Rules	163
3.3	Undertaking for Examination Rules and Regulations	164
3.4	Undertaking for Refraining from Possession and Use of Drugs and Alcohol	165
4	Important Contacts	166

PREAMBLE

The Handbook (Student's Information Booklet) for Students, printed in two volumes contain General Information Respectively about the CHARUSAT University and detailed information about Smt. Chandaben Mohanbhai Patel Institute of Computer Applications.

Handbook Volume-I contains information about general rules to be followed by the students on campus. It gives information about the general facilities and support available for the students on campus. It gives insight about the discipline and conduct rules of the University.

This Handbook (Student's Information Booklet) is for the purpose of providing information to the students about the University and its programmes and is not a Regulation book of the University. Hence, no claim can be made based on the information given in the book.

The University / Institute reserves the right to amend the rules and regulations mentioned in the Handbook without any prior notice. The decision of the University shall be final on all matters. For any clarification, the Student Section may be contacted. Handbook Volume-II (Student's Information Booklet) contains academic information about the Institute, which includes the Academic Rules and Regulations regarding academic requirements and academic conduct of the students at the University including different policies and forms. Besides, it includes important information on registration, grading system, academic standards, attendance norms, discipline and the like. The students shall abide by these rules and shall, at all times, conduct in a manner so as to bring credit to the University and enhance its prestige in the society.

It is prime responsibility of the students to get familiar (themselves) with the rules and regulations.

ABOUT INSTITUTE

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications (CMPICA) is a constituent institute of CHARUSAT. It is offering a MCA, M.Sc. (IT), BCA, B.Sc.(IT), Online BCA, Online MCA and Ph. D. programmes. The vision of the institute is to become one of the leading institutions in the country by imparting state-of-the-art education to the students in the field of computer applications and by contributing in the nation's efforts of computerizing public systems for the benefit of the masses. It is committed to excel in both teaching and research. Institute is reacting rapidly to the changing technological landscape by adapting quickly. The Institute is also committed to adapt industry practices.

The institute has received numerous prestigious awards from various organizations, including AICTE, the Government of India, and the Government of Gujarat. Additionally, it has been awarded a five-star rating in the Gujarat State Institutional Rating Framework (GSIRF) by the Government of Gujarat.

ACADEMIC REGULATIONS & SYLLABUS 2024 -25

Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute of
Computer Applications
Bachelor of Science in Information Technology
(BSc-IT) Programme
(as per NEP-2020)



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ACADEMIC RULES

To ensure uniform system of education, duration of under graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Bachelor's levels. Each semester will be at least 90 working days' duration. The Curriculum and Credit Framework for Undergraduate Programmes (CCFUP) is based on the University Grants Commission's (UGC) guidelines, with a student-centric focus, flexible choice-based credit system, and a multidisciplinary approach. This framework also provides multiple entry and exit options, facilitating students to align their education with their career aspirations by choosing subjects/fields of their interest. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Certification and Duration of Study with Multiple Entry Multiple Exit (MEME)

The undergraduate degree will have a flexible duration of either three or four years, with multiple exit options during this period. The table below outlines the various certifications a student can earn at different stages of their undergraduate study:

Table-1 Certification and Duration on MEME

Duration of Study	Semesters Completed	Certification Earned
4 Years	Eight Semesters(4 Academic Years)	Bachelor's Degree (Honours)
		Bachelor's Degree (Honours with Research)
3 Years	Six Semesters(3 Academic Years)	Bachelor's Degree
2 Years	Four Semesters(2 Academic Years)	Undergraduate Diploma
1 Year	Two Semesters(1 Academic Year)	Undergraduate Certificate

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Entry and Multiple Exit (MEME) scheme in their Bachelor of Science in Information Technology (BSc-IT) programme. This structure provides students with the flexibility to enter, exit, and re-enter the programme at various stages, each with its corresponding certification mentioned above in Table-1.

Table 2: Multiple Entry Multiple Exit options

NCrF Credit Levels	Qualification Title	Credit Requirement	Programme Exit	Additional Requirement	Re-entry to Degree Programme
4.5	UG Certificate	48	After Year 1	Completion of one 4- credit of summer internship in core specific NSQF defined course during summer vacation of Year 1	Within three years of exit
5	UG Diploma	92	After Year 2	Completion of one 4- credit of summer internship in core specific NSQF defined	Within three years of exit

				course during summer vacation of Year 2	
5.5	Three Year Bachelor Degree	132	After Year 3	Compliance with minimum credit requirements specified in regulations	Within three years of exit
6	Bachelor's Degree with Honors	176	After Year 4	Compliance with minimum credit requirements specified in regulations	NA
6	Bachelor's Degree with Honors with Research	176	After Year 4	Compliance with credit requirements as specified in regulations	NA

Students who are opting for exit at any level, shall re-enter the institution to complete the UG Degree, where they had left off. They can re-enter within three years of exit and complete the degree programme within the stipulated maximum period of seven years from the date of admission to UG programme.

3. CREDIT FRAMEWORK FOR BSc-IT(Bachelor of Science in Information Technology)

Arrangement of Credit Distribution Framework for three/four years Honors/Honors with Research Degree Programme with Multiple Entry and Exit Options.

(As per GR No: KCG/admin/2023-24/0607/kh.1, Sachivalaya, Gandhinagar, Date-11/07/2023).

Sr.No	Categories of Courses	Credit Requirement for each Category				
		Certificate (1 Year)	Diploma (2 Years)	3-Year UG	4-Year UG (Honors)	4-Year UG (Honors+Research)

1	Major - Core Courses	16	40	64	88	88
2	Minor-Discipline Specific Electives	8	12	24	32	32
3	Multidisciplinary Courses Open Electives	8	10	12	12	12
4	Ability Enhancement Courses(AEC)	4	8	10	10	10
5	Skill Enhancement Courses(SEC)	4	10	14	14	14
6	Value Added Courses (VAC)	4	8	8	8	8
7	Summer Internship/ Research Project /Dissertation	-	-	-	12	12
8	Exit Courses	4	4	-	-	
	Total	48	92	132	176	176

4. Structure of BSc-IT (Bachelor of Science in Information Technology)

NCrF Credit Level	Semester	Major Core	Minor	Multidisciplinary	Ability Enhancement Courses(AEC)	Skill Enhancement Courses(SEC)	Value Added Courses(VAC)/IKS	Research Project/Dissertation	Total	Qualification/Certificate
4.5 First Year	I	8	4	4	2	2	2	-	22	UG Certificate
	II	8	4	4	2	2	2	-	22	
1 st Year Credit Total		16	8	8	4	4	4	-	44	
Exit 1: Award of UG certificate in Major course with 44 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for the next NCrF credit level										
NCrF Credit Level	Semester	Major Core	Minor	Multidisciplinary	Ability Enhancement Courses(AEC)	Skill Enhancement Courses(SEC)	Value Added Courses(VAC)/IKS	Research Project/Dissertation	Total	Qualification/Certificate
5.0 Second Year	III	12	-	2	2	4	2	-	22	
	IV	12	4	2	-	2	2	-	22	

2 nd Year Credit Total	24	4	6	2	6	4	-	44	UG Diploma	
Grand Total 1 st and 2 nd Year Credit	40	12	12	6	10	8	-	88		
Exit 2: Award of UG Diploma in Major course with 88 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for t h e next NCrF credit level										
NCrF Credit Level	Semester	Major Core	Minor	Mul ti/I nte r Dis cipl inary	Ability Enhance ment Courses(AE C)	Skill Enhance ment Courses(SEC)	Value Added Courses(VA C)/IKS	Research Project/Diss ertation	Total	Qualification/ Certificate
5.5 Third Year	V	12	8	-	-	2	-	-	22	UG Degree
	VI	12	4	-	2	4	-	-	22	
3 rd Year Credit Total		24	12	-	2	6	-	-	44	
Grand Total 1 st to 3 rd Year Credit		64	24	12	08	16	8	-	132	
Exit 3: Award of UG Degree in Major course with 132 credits.										
NCrF Credit Level	Semester	Major Core	Minor	Mul ti/I nte r Dis cipl inary	Ability Enhance ment Courses(AE C)	Skill Enhance ment Courses(SEC)	Value Added Courses(VA C)/IKS	Research Project/Diss ertation	Total	Qualification/ Certificate
6 Fourth Year	VII	12	4	-	-	-	-	6(OJT)	22	UG Honors Degree
	VIII	12	4	-	-	-	-	6(OJT)	22	
4 th Year Credit Total		24	8	-	-	-	-	12	44	
Award of UG Honors Degree in Major course with 176 credits.										
6 Fourth Year	VII	12	4	-	-	-	-	6(RP)	22	UG Honors With Research Degree
	VIII	12	4	-	-	-	-	6(RP)	22	
4 th Year Credit Total		24	8	-	-	-	-	12	44	
Grand Total 1 st to 4 th Year Credit		88	32	12	08	16	8	12	176	
Award of UG Honors with Research Degree in Major course with 176 credits.										
*OJT – On the Job Training Courses Only										* RP – Research Project With Major Core

2 credit course on Community Service/NSS/NCC/Sports as a compulsory course to be undertaken by students during the initial two years of the study. This course will not affect the overall SGPA/CGPA of the students.

5. Eligibility & Mode of Admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

6. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

7. Attendance

7.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

7.2 Student attendance in a course should be 80%.

8. Course Evaluation

8.1 The performance of every student in each course will be evaluated as follows:

8.1.1 Continuous and Comprehensive Evaluation using various components such as Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc. by the course faculty member(s). The weightage of these components is 50% of the marks for the course; and

8.1.2 Semester-End-Evaluation by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 50% of the marks for the course.

8.2 University Examination

8.2.1 The final examination by the University for 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

8.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

8.2.3 Performance of Continuous and Comprehensive Evaluation(CCE) and Semester-End-Examination(SEE) Examination will be done on the grading system.

9. Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Grading Scheme Range of Marks (%)	96.0-100	86.0-95.9	76.0-85.9	66.0-75.9	56.0-65.9	46.0 – 55.9	36.0 – 45.9	Below 36.0	Absent
Letter Grade	O (Outstanding)	A+ (Excellent)	A (Very Good)	B+ (Good)	B (Above Average)	C (Average)	P (Pass)	F (Fail)	Ab (Absent)
Grade Point	10	9	8	7	6	5	4	0	0

$SGPA = \sum CiGi / \sum Ci$ where Ci is the number of credits of course i Gi is the Grade Point for the course i and $i = 1$ to n , n = number of courses in the semester.

$CGPA = \sum CiGi / \sum Ci$ where Ci is the number of credits of course i Gi is the Grade Point for the course and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed

The minimum passing marks for each pattern of evaluation are 36% .

10. Detention Rule

A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in. Awards of Degree.

10.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

10.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

10.1.2 He should have cleared all evaluation components in every course;

11. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥ 7 & ≤ 10
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First class:	CGPA ≥ 6.0 & <7
Second Class:	CGPA ≥ 5.0 & <6.0
Pass Class:	CGPA <5.0

12. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Choice of Courses

Undergraduate students will make course selections from the various lists and that will fall into categories as outlined below:

Major Course:

This is the primary subject area a student wishes to delve into during their undergraduate course. It is the area of study that the student is most passionate about and may wish to pursue as a career.

Minor Course:

This is a secondary area of study that complements the student's chosen Major. Students who complete a sufficient number of courses in a discipline or an interdisciplinary area of study other than their chosen Major will qualify for a Minor in that discipline or interdisciplinary area

Multidisciplinary Course:

All undergraduate students are required to complete three introductory-level courses relating to any of the broad disciplines. These courses aim to broaden the student's intellectual experience and form part of liberal arts and science education.

Ability Enhancement Course (AEC):

These are Modern Indian Language (MIL) & English language courses focused on enhancing language and communication skills. The courses aim to enable students to acquire and demonstrate core linguistic skills, including critical reading and expository and academic writing skills. The courses also emphasize the development and enhancement of skills such as communication, and the ability to participate in/conduct discussion and debate.

Skill Enhancement Courses (SEC):

These courses aim to impart practical skills, hands-on training, soft skills, etc., to enhance the employability of students. The institution may design courses as per the students' needs and available institutional resources.

Value Added Course (VAC):

These courses are designed to provide extra skills or knowledge beyond the standard curriculum, often tailored towards enhancing employability, promoting entrepreneurship, or developing personal and professional skills.

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Credit Hour Allocation for Different Course Types

Definition and Standardization

The workload associated with a course will be measured in terms of credit hours. One credit hour is equivalent to one hour of instructional time per week over the duration of a semester (minimum 15 weeks).

Types of Course Components

Courses can be comprised of one or more of the following components:

1. Lecture
2. Practicum
3. Seminar/Internship/ Studio Activities/Field Practice/ Projects / Engagement and Service.

Credit Hours per Course Component

The weekly and semester-wise instructional hours associated with each credit for different components are as follows:

Credit Hours per Course Component

Course Component	Weekly Hours per Credit	Total Hours in a 15-week Semester
Lecture	1 Hour	15 Hours
Practicum	2 Hours	30 Hours
Internship, Field Practice/Projects, Community Engagement and Service/NCC/NSS/Sports	2 Hours	30 Hours

Course Categories

Courses in the study program are categorised based on the nature of the learning activities. These include:

Lecture courses: These involve expert-led sessions related to a specific field.

Practicum or Laboratory work: These apply previously learned principles/theories in practical projects or lab activities.

Internship: These involve professional activities or work experiences, typically supervised by experts from the relevant external entities.

Field practice/projects: These involve field-based learning or projects supervised by external experts.

Community engagement and service/NCC/NSS/Sports: These involve activities exposing students to societal socio-economic issues, facilitating the application of theoretical and practical knowledge to real-life problems. It also involves skills and physical fitness among students through indoor & outdoor sports, field & track events.

Vision, Mission, PEOs, POs and PSOs

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates will:

PEO1: Be able to understand the requirement of computing problem and implement an effective solution.

PEO2: Be able to successfully take up various available career options.

PEO3: Be able to continuously learn in their preferred domains.

PEO4: Be able to acquit themselves in ethical and professional manner while demonstrating effective skills.

PROGRAM OUTCOMES (POs)

The Graduate of Computer Science and Applications will be able to:

PO1: Computational Knowledge: Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge to provide effective solution in the area of computing.

PO2: Problem Analysis: Identify, formulate, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

PO3: Design /Development of Solutions: Design and evaluate solutions for computing problems, and design and evaluate systems, components, or processes that meet specified needs by considering several societal aspects.

P04: Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools for any computing activities, with an understanding of the limitations.

P05: Professional Ethics: Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

P06: Life-long Learning: Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

P07: Project management: Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

P08: Communication Efficacy: Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

P09: Societal and Environmental Concern: Understand and assess societal and environmental issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.

P10: Individual and Team Work: Function effectively as an individual and as a member or a leader in diverse teams and in multidisciplinary environments.

P11: Innovation and Entrepreneurship: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

PROGRAM SPECIFIC OUTCOMES

At the end of the programme, the student should be able to

PSO1: Analyze, evaluate and formulate workable computing solutions to real world problems using computing methods.

PSO2: Apply proficiencies of communication, teamwork and management skills with spirit of entrepreneurship.

SALIENT FEATURES

- Four Years unique programme as per National Education Policy-2020.
- Have a flexible duration of either three or four years, with multiple exit options during this period.
- Provides students with the flexibility to enter, exit, and re-enter the programme at various stages, each with its corresponding certification.
- Award of Degree:
 - **UG Certificate** (Candidate wish to Exit after the First Year with additional requirements of 4 credits summer internship)
 - **UG Diploma** (Candidate wish to Exit after the Second Year additional requirements of 4 credits summer internship)
 - **UG Degree** (Candidate wish to Exit after the Third Year)
 - **UG Honors Degree or UG Honors with Research Degree** (Candidate finishes all Four Years)
- Programme is designed with various categories of courses such as Major, Minor, Multidisciplinary, Ability Enhancement, Skill Enhancement, Value Added, Community engagement and service/NCC/NSS/Sports based on Choice Based Credit System(CBCS).
- Focus on student-centered learning.
- Problem-based learning and industry-relevant curriculum.
- Industry visits, seminars, workshops, and guest lectures by eminent researchers and industry practitioners.
- Active student clubs and bodies.
- The multi-cultural and vibrant teaching environment.

Total Credits of Programme (8 Semesters): 176

**TEACHING SCHEME AND
EXAMINATION SCHEME

FOR

B.Sc. (IT) PROGRAMME

EFFECTIVE FROM
ACADEMIC YEAR 2024-25**

Credit Framework for 3 Years/4 Years UG Programme

NCrF Credit Levels	Qualification Title	Credit Requirement	No. of Semesters	Year
4.5	UG Certificate	48	2	1
5	UG Diploma	92	4	2
5.5	Three Year Bachelor Degree	132	6	3
6	Bachelor's Degree with Honors OR Bachelor's Degree with Honors with Research	176	8	4
• 1 credit = 1 Hour of Theory • 1 credit = 2 Hour of Practical/Project				

Degree programs offered by Faculty

- Bachelor of Science in Information Technology (Honors) / Bachelor of Science in Information Technology (Honors with Research) (4-Year Programme) and maximum duration of the programme is 7 Years.

CREDIT FRAMEWORK For B.Sc. (IT)
(Bachelor of Computer Science in Information Technology)

Arrangement of Credit Distribution Framework for three/four years Honors/Honors with Research Degree Programme with Multiple Entry and Exit Options

(As per GR No: KCG/admin/2023-24/0607/kh.1, Sachivalaya, Gandhinagar, Date-11/07/2023)

Sr. No.	Categories of Courses	Credit Requirement for each Category				
		1 Year (Certificate)	2 Years (Diploma)	3-Year (UG)	4-Year UG (Honors)	4-Year UG (Honors + Research)
1	Major - Core Courses	16	40	64	88	88
2	Minor-Discipline Specific Electives	8	12	24	32	32
3	Multidisciplinary Courses Open Electives	8	12	12	12	12
4	Ability Enhancement Courses (AEC)	4	8	10	10	10
5	Skill Enhancement Courses (SEC)	4	8	14	14	14
6	Value Added Courses (VAC)	4	8	8	8	8
7	Summer Internship/ Research Project /Dissertation	-	-	-	12	12
8	Exit Courses	4	4	-	-	
	Total	48	92	132	176	176

Structure of B.Sc. (IT) (Bachelor of Computer Science in Information Technology)

NCrF Credit Level	Semester	Major Core	Minor	Multi / Inter Disciplinary	Ability Enhancements Courses (AEC)	Skill Enhancement Courses (SEC)	Value Added Courses (VAC) / IKS	Research Project / Dissertation	Total	Qualification / Certificate
4.5 First Year	I	8	4	4	2	2	2	-	22	UG Certificate
	II	8	4	4	2	2	2	-	22	
1 st Year Credit Total		16	8	8	4	4	4	-	44	
Exit 1: Award of UG certificate in Major course with 44 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for the next NCrF credit level										
NCrF Credit Level	Semester	Major Core	Minor	Multi / Inter Disciplinary	Ability Enhancements Courses (AEC)	Skill Enhancement Courses (SEC)	Value Added Courses (VAC)/IKS	Research Project / Dissertation	Total	Qualification /Certificate
5.0 Second Year	III	12	-	2	2	4	2	-	22	UG Diploma
	IV	12	4	2	2	-	2	-	22	
2 nd Year Credit Total		24	4	4	4	4	4	-	44	
Grand Total 1 st and 2 nd Year Credit		40	12	12	8	8	8	-	88	
Exit 2: Award of UG Diploma in Major course with 88 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for the next NCrF credit level										
NCrF Credit Level	Semester	Major Core	Minor	Multi/Inter Disciplinary	Ability Enhancements Courses (AEC)	Skill Enhancement Courses (SEC)	Value Added Courses (VAC)/IKS	Research Project / Dissertation	Total	Qualification /Certificate
5.5 Third Year	V	12	8	-	-	2	-	-	22	UG Degree
	VI	12	4	-	2	4	-	-	22	
3 rd Year Credit Total		24	12	-	2	6	-	-	44	
Grand Total 1 st to 3 rd Year Credit		64	24	12	10	14	8	-	132	
Exit 3: Award of UG Degree in Major course with 132 credits.										
NCrF Credit Level	Semester	Major Core	Minor	Multi/Inter Disciplinary	Ability Enhancements Courses (AEC)	Skill Enhancement Courses (SEC)	Value Added Courses (VAC)/IKS	Research Project / Dissertation	Total	Qualification / Certificate
6 Fourth Year	VII	12	4	-	-	-	-	6 (OJT)	22	UG Honors Degree
	VIII	12	4	-	-	-	-	6(OJT)	22	
4 th Year Credit Total		24	8	-	-	-	-	12	44	
Award of UG Honors Degree in Major course with 176 credits.										
6 Fourth Year	VII	12	4	-	-	-	-	6 (RP)	22	UG Honors With Research Degree
	VIII	12	4	-	-	-	-	6 (RP)	22	
4 th Year Credit Total		24	8	-	-	-	-	12	44	
Grand Total 1 st to 4 th Year Credit		88	32	12	10	14	8	12	176	
Award of UG Honors with Research Degree in Major course with 176 credits.										
*OJT – On the Job Training * RP – Research Project With Major Core Courses Only										

**TEACHING SCHEME
&
EXAMINATION SCHEME

FOR

B.Sc. (IT) PROGRAMME

EFFECTIVE FROM
ACADEMIC YEAR 2024-25**

B.Sc. (IT) Semester-I

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC101	Introduction to Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC102	Relational Database Administration	Practical	-	-	25	25	02	Major Core
CTUE101 CTUE102 CTME101 CTME102	(Select any one) 1. Fundamentals of Business Organization 2. Information Technology in Business 3. Computer Application in Business 4. Basics of Digital Marketing	Theory	50	50	-	-	04	Minor
CTUD101	Fundamentals of Digital Electronics	Theory	50	50	-	-	04	MDC
CTUS101	Mathematics for Computer Science	Theory	25	25	-	-	02	SEC
CLUV101	Environmental Sciences	Practical	-	-	25	25	02	VAC
HSUA101	Communicative English	Practical	-	-	25	25	02	AEC
CUUV101 CUUV102	(Select any one) Community Engagement and Sustainable Development Physical Education and Sports	Practical	-	-	25	25	02	VAC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

B.Sc. (IT) Semester-II

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC103	Introduction to Object Oriented Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC104	Fundamentals of Web Designing	Practical	-	-	25	25	02	Major Core
CTUE103 CTUE104	(Select any one) 1. Business Models for e-Commerce 2. Managerial Economics	Theory	50	50	-	-	04	Minor
CTUD102	Data Communication and Networks	Theory	50	50	-	-	04	MDC
CTUV101	Fundamentals of Accounting	Theory	25	25	-	-	02	VAC
HSUS101-117	(Select any one) A course on Liberal Arts HSUS101 – Painting HSUS102– Photography HSUS103– Sculpting HSUS104– Pottery and Ceramic Art HSUS105-Media and Graphic Design HSUS106-Art and Craft HSUS109- Dramatics HSUS110-Contemporary Dance HSUS111-Music (Vocal) HSUS112-Music (Instrumental)-Tabla HSUS113-Music (Instrumental)-Guitar HSUS114-Music (Instrumental)-Harmonium HSUS115-Music (Instrumental)- Flute HSUS116- Indian Classical Dance-Kathak HSUS116- Indian Classical Dance-Bharatnatyam	Practical	-	-	25	25	02	SEC
CTUA101	Presentation Skills	Practical	-	-	25	25	02	AEC
CTUI101	Summer Internship and Viva - I	VIVA	-	-	50	50	4	Exit Course

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
*SEE- Semester –End- Evaluation

B.Sc. (IT) Semester-III

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC201	Introduction to Data Structures	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC202	System Analysis and Design	Theory	50	50	-	-	04	Major Core
CTUC203	Programming the Internet	Practical	-	-	25	25	02	Major Core
CTUS201	(Select any one) 1. Computer Oriented Numerical Methods	Theory	50	50	-	-	04	SEC
CTUS202	2. Computer Oriented Management System							
CTUA201	Life Management	Theory	25	25	-	-	02	AEC
	University Elective-I	Practical	-	-	50	50	02	MDC
HSUV201	Creativity, Problem Solving and Innovation	Practical	-	-	25	25	02	VAC

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)

*SEE- Semester –End- Evaluation

University Elective - I			
Sr. No.	Course Code	Course Name	Department/Faculty
1		Art of Programming	Engineering
2		Environmental Sustainability & Climate Change	Engineering
3		ICT Resources & Multimedia	Engineering
4		Engineering Drawing	Engineering
5		Fundamentals of Packaging	Pharmacy
6		Basic Laboratory Techniques	Applied Science
7		First Aid & Life Support	Nursing
8		Health Promotion & Fitness	Physiotherapy
9		Banking & Insurance	Management
10		Introduction to MATLAB Programming	Engineering
11		Astrophysics, Space and Cosmos-1	Applied Science
12		SDG Handprint Laboratory	Engineering
13		Python Programming	Engineering
14	CAUD201	Introduction to Web Designing	Computer Science

B.Sc. (IT) Semester-IV

Course Code	Course Name	Theory / Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC204	Fundamentals of Operating Systems	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC205	Fundamentals of Visual Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUE201	(Select any one) 1. Introduction to Artificial Intelligence	Theory	50	50	-	-	04	Minor
CTUE202	2. Computer Network Security							
CTUA202	Social Media and Blog Writing	Practical	-	-	25	25	02	AEC
	University Elective-II	Practical	-	-	50	50	02	MDC
HSUV203	Indian Knowledge System	Practical	-	-	25	25	02	VAC
CTUI201	Summer Internship and Viva - II	VIVA	-	-	50	50	04	Exit Course
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)								
*SEE- Semester –End- Evaluation								

University Elective - II			
Sr. No.	Course Code	Course Name	Department/Faculty
1		Prototyping Electronics with Arduino	Engineering
2		Web Designing	Engineering
3		Basics of Environmental Impact Assessment	Engineering
4		Internet Technology & Web Design	Engineering
5		Material Science	Engineering
6		Cosmetics in Daily Life	Pharmacy
7		Life Style Diseases & Management	Nursing
8		Occupational Health & Ergonomics	Physiotherapy
9		Health Care Management	Management
10		Astrophysics, Space and Cosmos-2	Applied Science
11		MATLAB Programming	Engineering
12		Maintenance of Household Apparatus	Engineering
13	CAUD202	Programming the Internet	Computer Science

B.Sc. (IT) Semester-V

Course Code	Course Name	Theory / Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC301	Object Oriented Programming Using JAVA	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC302	Open Source Technology	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUE301 CTUE302	(Select any one) 1. Data Science Essentials 2. Basics of Digital Image Processing	Theory	25	25	-	-	02	Minor
CTUE303	Object Oriented Analysis and Design	Theory	50	50	-	-	04	Minor
CTUE304 CTUE305	1. Advanced Internet of Things 2. Introduction to Game Development	Practical	-	-	25	25	02	Minor
HSUS301	French	Practical	-	-	25	25	02	SEC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

B.Sc. (IT) Semester-VI

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC303	Project Work	VIVA	150	150	-	-	12	Major Core
CTUE306 CTUE307	(Select any one) 1. Multi-paradigm Programming Language 2. Frameworks and Applications	Theory and Practical	25	25	25	25	04 (02-Theory) (02-Practical)	Minor
CTUS301	Mobile Application Development	Theory and Practical	25	25	25	25	04 (02-Theory) (02-Practical)	SEC
HSUA302	Professional Communication, Soft Skills and Personality Development	Practical	-	-	25	25	02	AEC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

B.Sc. (IT) Semester-VII

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC401	Enterprise Computing using Java EE	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC402	Database Technologies	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUE401 CTUE402 CTUE403	(Select any one) 1. Cloud Computing 2. Cryptography & Network Security 3. Blockchain Essentials	Theory	50	50	-	-	04	Minor
CTUP401 CTUR401	(Select anyone) 1. On-Job Training-I 2. Research Project-I	VIVA	-	-	75	75	06	Research Project / Dissertation
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

B.Sc. (IT) Semester-VIII

Course Code	Course Name	Theory / Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC403	Full Stack Web Development	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC404	Advanced Mobile Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUE404	Web Development using Open Source Technologies	Practical	-	-	25	25	2	Minor
CTUE405 CTUE406 CTUE407	(Select any one) 1. Python Web framework 2. HTTP Web Service for Enterprise Applications 3. Game Development using Unity	Practical	-	-	25	25	2	Minor
CTUP402 CTUR402	(Select any one) On Job Training-II Research Project-II	VIVA	-	-	75	75	6	Research Project / Dissertation
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)								
*SEE- Semester –End- Evaluation								

**TEACHING SCHEME &
DETAILED SYLLABUS**

FOR

**B.Sc. (IT) PROGRAMME
(1st SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2024-25**

B.Sc. (IT) Semester-I

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC101	Introduction to Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC102	Relational Database Administration	Practical	-	-	25	25	02	Major Core
CTUE101	(Select any one) 1. Fundamentals of Business Organization	Theory	50	50	-	-	04	Minor
CTUE102	2. Information Technology in Business							
CTME101	3. Computer Application in Business							
CTME102	4. Basics of Digital Marketing							
CTUD101	Fundamentals of Digital Electronics	Theory	50	50	-	-	04	MDC
CTUS101	Mathematics for Computer Science	Theory	25	25	-	-	02	SEC
CLUV101	Environmental Sciences	Practical	-	-	25	25	02	VAC
HSUA101	Communicative English	Practical	-	-	25	25	02	AEC
CUUV101	(Select any one) Community Engagement and Sustainable Development	Practical	-	-	25	25	02	VAC
CUUV102	Physical Education and Sports							
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)								
*SEE- Semester –End- Evaluation								

CTUC101 - Introduction to Programming

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory) : Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical) : Lab Assignments, Practical Test, VIVA					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures, facilitate problem-solving skills by employing algorithms and flowcharts, enabling the understanding of the C programming language's structural aspects, and fostering the capability to create basic programs incorporating data types, variables, and constants. Expand comprehension of diverse conditional and iterative statements and expressions, while also delving into the manipulation of arrays encompassing both numerical and character elements. Develop proficiency in comprehending user-defined functions and explore advanced programming concepts such as utilizing arrays with pointers and structures promoting independent learning and application in real-world scenarios.

Outline of the course (Theory):

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Programming Basics	08
2	Introduction to C	12
3	Expressions and Control Statements	10
4	Arrays and Strings	10
5	Introduction to Functions	10
6	Pointers & Basic of the structures	10

Total Hours (Theory): 60

Total hours (Practical) : 60

Total Contact Hours : 120

Detail Syllabus:

Unit I: Programming Basics

Hours 08

Types of Programming languages and its History, features and application. Introduction to algorithm, Key features of algorithm, Introduction to flow charts, Significance of flow chart, Advantages and limitation of flow chart, Introduction to programming languages, Introduction to editor, compiler, and translator

Unit II: Introduction to C

Hours 12

Introduction, characteristics of C, Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program. Basic data types in C, User define data types, Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C, Operators - arithmetic, relational, logical, assignment, increment - decrement, conditional, Bitwise, comma operator, size-of operator, operator precedence chart.

Unit III: Expressions and Control Statements

Hours 10

Arithmetic expressions, evaluation of expressions, type conversions in expressions, operator precedence and associativity, mathematical functions. Type conversion and casting; Introduction to decision control statements, Conditional and branching statements, loop, nested loop, Break, continue and go to statement.

Unit IV: Arrays and Strings

Hours 10

Arrays: One-dimensional, two-dimensional, Handling of Character Strings: Declaring and initializing string variables, reading string from terminal, writing string to screen, string handling functions, table of strings.

Unit V: Introduction to Functions

Hours 10

Need for user-defined functions, the form of c function, return values and their types, calling a function, category of functions, handling of non-integer functions, nesting of Functions, recursion, functions with arrays.

Unit VI Pointers & Basic of the structures

Hours 10

Understanding the Computer's Memory, Introduction to Pointers, Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic, Null Pointers, Generic Pointers, passing arguments to Function using Pointers, Pointers and Arrays, Passing an Array to a Function, Difference between Array name and Pointer, Pointers and Strings, Array of Pointers. Defining a structure, declaring structure variables, accessing structure members, structure initialization,

Outline of the Course(Practical) :

Week No	Practical	Description
1	Introduction to Programming and IDE	Overview of programming languages, algorithms, flowchart, Setting up the development environment, Introduction of development environment.
2-3	Basic of C programming	Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program; Basic data types in C. Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C
4-5	Operators, Expressions evaluation & type conversions.	Operators - arithmetic, relational, logical, assignment, increment - decrement, conditional, Bitwise and special, comma operator, size-of operator, operator precedence chart. Arithmetic expressions, evaluation of expressions, type conversions.
6-7	Decision-making statements and looping statements	Decision-making statements (if, else if, else) and nested decision-making statements, Switch statement, Looping statements (for, while, do-while), nested looping statements Break and continue statements.
8-9	Arrays	Understanding arrays, one-dimensional array, two-dimensional arrays, practical exercises of one-dimensional and two-dimensional array.
10-11	String and String functions	Declaring and initializing string variables, reading string from terminal, writing string to screen, string-handling functions.
12-13	User defined functions	Introduction to functions, Function declaration, definition, and calling, Parameters and return values. Scope and lifetime of variables, Recursion.
14-15	Pointers & Structures	Introduction to pointers, Pointers and arrays, Pointer arithmetic, Defining and declaring structures, Accessing structure members

Total hours: 60**Core Books:**

1. ReemaThareja: Computer Fundamentals and Programming in C, 2nd Edition, Oxford University press, 2017

2. PradipDey and Manas Ghosh. "Programming in C", Second Edition, Oxford University Press.
3. E. Balagurusamy: Programming In C, 7th Edition.
4. Asok N Kamthane, Pearson, Programming in C
5. Anita Goel, Pearson, Computer Fundamentals
6. Schaum Series, Gottfried B.S., Tata McGraw Hill, Programming with C

Reference Books:

1. Brian W. Kerighan and Dennis M. Ritchie : The C Programming Language, Prentice Hall.
2. PradipDey, Manas Ghosh: Programming in C, 4th Edition, Oxford University press.
3. Herbert Schildt: The complete reference C, Fourth edition, Tata McGraw Hill
4. Yashwant Kanetkar: Let us C, 13th Edition, BPB publication.
5. Rajaraman V, PHI, Computer Basics and Programming in C

Web References:

1. <https://cprogrampracticals.blogspot.com/p/basic-concepts.html> [For basic c programming practical]
2. practical]
3. <https://www.programtopia.net/c-programming/docs/operators-expressions> [For operators and expressions in C]
4. <https://www.javatpoint.com/functions-in-c> [For functions]
5. <https://www.programiz.com/c-programming> [For basic c programming practical]
6. <https://www.javatpoint.com/c-pointers> [pointers in c]
7. <https://www.programiz.com/c-programming/c-structures> [structures in c]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be able to solve problems using algorithms and flowcharts.
CO2 :	Learn about the structure of C program and able to develop simple programs using the C Programming language with data types, variables and constants.
CO3 :	Gain the knowledge of different conditional and iterative statements and expressions.
CO4 :	Explore the utilization and manipulation of numbers and characters arrays both.
CO5 :	Be able to understand user define function. Also, learn Programing-using Array with Pointer and Structure.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Programming Basics	√				
2	Introduction to C		√			
3	Expressions and Control Statements			√		
4	Arrays and Strings		√	√	√	
5	Introduction to Functions		√	√		√
6	Pointers & Basic of the structures		√			√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUC103 - Relational Database Administration

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	02	60	40	100
Experiential Learning Components: Lab Assignments, Practical Test, VIVA , certification courses, development of projects etc.					

Prerequisites: Basic knowledge of working with computer

Methodology, Pedagogy & Andragogy: Learn about computer data. Understand what data is, learn its meaning, examine the types of data sources, and see the difference between data and information. The database applications used in the real world will be discussed with necessary examples. During the laboratory hours' students' will implement the concepts. Apply SQL commands to mine data from databases and data visualization techniques to address organization information needs.

A. Outline of the Course:

Week No	Practical	Description
1	Introduction to Database System	Overview of data, information, file, database, database systems, DBMS, Purpose of DBMS over file system.
2-3	Data modelling and architecture	Various data models: ER model and Relational model, Three level architecture, structure of DBMS and database actors and workers, roles of database administrator
4-5	Database Design Methodology	CODD Rules, Functional Dependency and Normalization for Database Informal Design Guidelines for Relational Schemas, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF)
6-7	Schema Definition, Constraints, Queries – I	Basic Data types, Create Table Command, Modifying the structure of tables, renaming table, truncating table, destroying table, Insert, Delete and Update Statements in SQL

8	Schema Definition, Constraints, Queries – II	Data Constraints and Functions: - Pseudo columns, Null values, TAB table, DUAL table Operators, Data constraints, Type of data constraints, modifying constraints, working with data dictionary and use of USER_CONSTRAINTS Functions introduction
9	Inbuilt functions	Merits and demerits of functions, Types of functions: Numeric functions, Character functions, Date functions, Conversion functions, Aggregate functions
10-11	Querying the data	Different operators: LOGICAL, range searching and pattern matching, Viewing Data in the tables, sorting
12-13	Advanced Query Processing-I	Co-related Nested Queries, Group By clause, having clause, Data Control Language Commands: GRANT and REVOKE
14-15	Advanced Query Processing-II	Joins (Inner Join, Outer Join, Self-Join, Equi Join, Cross Join), Creation and manipulation of database objects indexes, views, sequences and synonym.

Total hours: 60

Text Books:

1. Ivan Bayross : SQL, PL/SQL The programming Language Oracle (4th Revised edition).
2. Ramkrishnan, Gehrke : Database Management Systems, 3rd Edition, McGrawHill Publication.
3. RamezElmasri, Shamkant B. Navathe : Fundamentals of Database Systems , 5th Edition, , Pearson Publication.

Reference Books:

1. Silberschatz, Korth, Sudarshan : Database System Concepts, 5th Edition, McGraw Hill.
2. C.J.Date, a Kannan, S Swaminathan : An Introduction to Database Systems, 8th Edition, Pearson Education,(Equivalent Reading).
3. Scoot Urban : Oracle 9i, PL/SQL Programming, Oracle Press.
4. S. K. Singh : Database Systems: Concepts, Design and Applications, Pearson Education
5. Peter Rob, Carlos Coronel: Database Systems: Design, Implementation and Management, 7th Edition, Cengage Learning, 2007.

6. Anjali Jivani and Amisha Shingala, "Practice book on SQL and PL/SQL with solutions", Roopal and Nirav publication.
7. Leon and Leon : Database management Systems, Vikas Publication.

Web References:

1. <https://www.javatpoint.com/er-model-vs-relational-model> [For difference of ER and Relationmodel]
2. http://www.microsoftvirtualacademy.com/trainingcourses/databasefundamentals#?fbid=tbZ92pOp_Tt [For overall subject]
3. http://www.ntu.edu.sg/home/ehchua/programming/sql/Relational_Database_Design.html [for relational database design]
4. http://docs.oracle.com/cd/A97335_02/apps.102/a81358/05_dev1.htm [For ER Diagram]
5. <http://plsql-tutorial.com/> [For PL/SQL]

Course Outcome: Upon successful completion of the course, student will:

C01:	Understand basic concepts regarding data, database systems and various data models.
C02:	Understand various data modeling techniques and entity relationship models.
C03:	Learn about relational data model and related concepts.
C04:	Gain insights into database design, SQL and normalization concepts.
C05:	Advanced query processing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PS01	PS02
C01	2	3	1	1	-	3	-	-	-	-	-	2	-
C02	1	3	3	3	-	3	1	-	-	-	-	1	3
C03	3	3	3	3	-	3	2	-	-	2	-	2	
C04	-	3	3	3	-	3	3	-	-	3	3	-	3
C05	2	3	3	3	2	3	3	-	-	-	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CTUE101 - Fundamentals of Business Organization

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the Fundamentals of Business Organization. Students will be introduced to Business Fundamentals, Business Enterprises, Business Services, Emerging Trends of Business, Managing Financial Resources, and Business in Global Environment. Students will give practical exposure in form of case study and by visit to E-commerce / Digital Marketing company and/or supply chain Centre.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Business Fundamentals	09
2	Business Enterprises	11
3	Business Services	09
4	Emerging Trends of Business	11
5	Managing Financial Resources	09
6	Business in Global Environment	11

Total Hours (Theory): 60

Total Contact Hours: 60

Detail Syllabus:

Unit I : Foundation of Information Technology

Hours 09

Introduction to Information Technology, Acquisition of various types of data- Introduction to Number and Textual Data, Image Data, Audio and Video Data Numbers and Textual Data, Data Storage and Organization

Unit II : Business Fundamentals

Hours 11

Business – Concept, nature and scope, business as a system, business objectives, business and environment interface, distinction between business, commerce and trade.

Unit III : Business Enterprise

Hours 09

Forms of business organization: Sole proprietorship, Joint Hindu Family Firm, partnership firm, joint stock Company, types of cooperative society: Limited Liability Partnership, Choice of form of organization, multinational corporations.

Unit IV : Emerging Trends of Business

Hours 11

E – business – Meaning, Scope and benefits, Resource required for successful E –Business implementation, on – line transactions, payment mechanism, Security and safety of business transactions, Outsourcing – Concept, need and scope

Unit V : Managing Financial Resources

Hours 09

Functions of money, financial institutions, role of financial manager, understanding security markets, career in finance.

Unit VI : Business in Global Environment

Hours 11

Globalization of business, threats and opportunities in global business, global business environment, trade controls, reducing International trade barriers, career in international business

Core Books:

1. Koontz: Principles Of Management (Ascent Series) Paperback: Tata McGraw Hill Education, 2004.
2. Y.K. Bhushan: Fundamentals of Business Organization, Sultan Chand Publisher, 2018

Reference Books:

1. Dr. Shveta Klara, Dr. Neha Singhal: Business Organization and Management, Scholar Tech Press, 2020.
2. Karen M. Collins, Jacqueline Shemko: Exploring Business, Pearson Education, 2008.

3. Stephen J. Skripak : Fundamentals of Business : 3rd Edition, Virginia Tech Libraries, 2018.

Web References:

1. [https://vtechworks.lib.vt.edu/bitstream/handle/10919/70961/Chapter%204%20Business %20in%20a%20Global%20Environment.pdf](https://vtechworks.lib.vt.edu/bitstream/handle/10919/70961/Chapter%204%20Business%20in%20a%20Global%20Environment.pdf) [Business in Global Environment]
2. <https://www.managementstudyguide.com/financial-management.htm> [Managing Financial Resources]
3. <https://leverageedu.com/blog/emerging-modes-of-business/> [Emerging Trends of Business]
4. <https://open.umn.edu/opentextbooks/textbooks/319> [ebook for reference]
5. <http://thuvienso.bvu.edu.vn/bitstream/TVDHBRVT/15850/1/An-Introduction-to-Business-and-Business-Planning.pdf> [Introduction to Business and Business Planning]

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	Able to understand business fundamentals
C02 :	Able to understand different types of business proprietorship.
C03 :	Able to get the knowledge of functionality of various online business services.
C04 :	Able to understand the basics to manage finance for business
C05 :	Learned the concepts and the scope of international business enterprise.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Business Fundamentals	√				
2	Business Enterprises		√			
3	Business Services			√		
4	Emerging Trends of Business			√		
5	Managing Financial Resources				√	
6	Business in Global Environment					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	-	-	-	-	-	3	-	-	-	-	-	-	1
C02	-	-	-	-	-	3	-	-	-	-	2	-	2
C03	-	2	2	2	-	3	2	-	-	2	2	-	2
C04	-	-	-	-	-	3	-	-	-	-	-	-	2
C05	-	-	-	-	-	3	-	-	1	1	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUE102 - Information Technology in Business

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the Information Technology in Business. Students will be introduced to Foundation of Information Technology, Internet Applications, Internet Service Trends in Business, E-Business, Data Analytics, Modern Business Management, Emerging Trends, and Societal Impact of IT. Students will give practical exposure in form of case study and by visit to E-commerce / Digital Marketing company and/or supply chain Centre.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Foundation of Information Technology	09
2	Internet Applications	11
3	Internet Service Trends in Business	09
4	E-Business and Data Analytics	11
5	Modern Business Management	09
6	Emerging Trends and Societal Impact of IT	11

Total Hours (Theory): 60

Total Contact Hours: 60

Detail Syllabus:

Unit I : Foundation of Information Technology

Hours 09

Introduction to Information Technology, Acquisition of various types of data- Introduction to Number and Textual Data, Image Data, Audio and Video Data Numbers and Textual Data, Data Storage and Organization

Unit II : Internet Applications

Hours 11

Internet: Introduction, Internet basics, Internet protocols, Internet addressing, Browser WWW, E-mail, telnet, FTP, application, benefits and limitation of internet, electronic conferencing, and teleconferencing.

Unit III : Internet Service Trends in Business

Hours 09

Fundamentals of Cloud Computing, Types of Cloud based Services, Varieties of Cloud Computing, Types of Cloud Computing, Advantages and Disadvantage of Cloud Computing, Cloud Computing Providers, Application of Cloud Computing in Business Area. E-commerce, E-Marketing, E-Branding, E-Advertising.

Unit IV : E-Business and Data Analytics

Hours 11

E-Business: Introduction, Components, Difference between E-Commerce and E-Business ; Types of Data – Qualitative and Quantitative, Data Analytics basics, Types of Analytics, Common technologies used in data analytics, Tools for Data Analytics

Unit V : Modern Business Management

Hours 09

E-SCM: Introduction, Supply Chain Management, E- Supply Chain Management, Component of Modern E-SCM Major Trends in E-SCM, Example of E-SCM, Architecture of E-Supply Chain Models

E-CRM: Customer Relationship Management Concept, E-CRM Solution Advantages of E-CRM Solutions, Advantages of E-CRM, E-CRM Capabilities, Example of E-CRM, E-CRM Framework

Unit VI : Emerging Trends and Societal Impact of IT

Hours 11

Internet of Things, Virtual and Augmented Reality, Industry 4.0, Image Processing; Introduction to Societal Impact, Social Use of World Wide Web, Privacy, Security and Integration of Information, Disaster Recovery, Intellectual Property Right, Career in IT

Core Books:

1. V Rajaraman : Introduction to Information Technology, 2nd Edition, PHI Learning Private Limited, 2013.
2. P.T. Joseph, S.J.: E-Commerce An Indian Perspective, 5th Edition, PHI Learning Private Limited, 2015.
3. R. Kalakota & M. Robinson, e-Business 2.0, 2nd Edition, Pearson Publisher, 2012.
4. Thomas Erl, Z. Mahmood and R. Puttini: Cloud Computing Concepts, Technology & Architecture, Pearson Publication, 2014

Reference Books:

1. Syed Muhammad Fahad Akhtar, Big Data Architect's Handbook, Packt Publishing Ltd, 2018.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education, 2014.

Web References:

1. <https://blog.k2datascience.com/the-basics-of-data-analytics-77e5cc7ea741> [For Data Analytics]
2. <https://www.investopedia.com/terms/c/cloud-computing.asp> [For Cloud Computing]
3. <https://www.javatpoint.com/cloud-computing-applications> [For Cloud Computing]
4. <https://www.geeksforgeeks.org/difference-between-e-commerce-and-e-business/> [For E-business vs E-Commerce]
5. <https://www.temok.com/blog/what-is-e-business/> [For E- business]
6. <https://files.eric.ed.gov/fulltext/ED536788.pdf> [For Data Analytics]
7. <https://vdocument.in/introduction-to-e-scm.html> [For E-SCM]
8. <http://nek.istanbul.edu.tr:4444/ekos/TEZ/42370.pdf> [For E-CRM]
9. https://wps.prenhall.com/wps/media/objects/10704/10961611/Online_Appendix_B.pdf [For E-CRM]

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	Understand the basics of Information Technology.
C02 :	Aware about Internet based Applications for any business context.
C03 :	Understand trends of Internet services in Business.
C04 :	Understands basics of Data Analytics and E-business concepts.
C05 :	Understand modern business management concepts, emerging trends and societal Impact of Information Technology.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Foundation of Information Technology	√				
2	Internet Applications		√	√		
3	Internet Service Trends in Business			√	√	
4	E-Business and Data Analytics				√	
5	Modern Business Management					√
6	Emerging Trends and Societal Impact of IT					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	3	1	1	-	-	2	-	-	-	-	-	2	2
C02	3	1	2	1	-	3	-	-	-	-	2	2	2
C03	3	1	2	1	-	3	-	-	-	-	2	2	2
C04	3	1	2	1	-	3	-	-	-	-	2	2	2
C05	3	1	2	1	2	2	-	-	2	-	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTME101: Computer Application in Business

Description:

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The main objective of this course is to familiarize the students with the application of computer in business and how it facilitates in business decision making.

Pre-requisites:

Basic knowledge of computers

Course layout:

Weeks	Video No.	Videos Title	Units
Week-1	Video-1	Introduction to Computers	Unit-1
	Video-2	Application in Computers	Unit-2
	Video-3	Web Application	Unit-3
	Video-4	Basic of Computer Software	Unit-4
Week-2	Video-5	Business Information System-	Unit-5

	Video-6	Database Management System	Unit-5
	Video-7	Business Process & Enterprises Application	Unit-5
	Video-8	GDPR	Unit-5
Week-3	Video-9	IT Securities Measures in Business	Unit-6
	Video-10	Internet Services and Email	Unit-7
	Video-11	Plastic Money, E-Wallet and Online Pay	Unit-8
Week-4	Video-12	Basics of Word Processing	Unit-9 & Unit-10
	Video-13	Tools using Word	Unit-11
	Video-14	Advance Tool using Word Processing	Unit-11
	Video-15	Making Business Documentation using word	Unit-12
Week-5	Video-16	Working with PowerPoint	Unit-13
	Video-17	PPT usages in Business and Corporate	Unit-13

Week-6	Video-18	Meaning of Multimedia	Unit-14
	Video-19	Animation & Presentation	Unit-14
	Video-20	Implementing multimedia tool in business	Unit-14
	Video-21	Meaning and Role of You Tube in Business	Unit-15
	Video-22	Video Impact in Excelling Business	Unit-15
Week-7	Video-23	Curtain raiser to Spreadsheet & its Utility in Business World	Unit-16
	Video-24	Spreadsheet concepts-1	Unit-16
	Video-25	Spreadsheet concepts-II	Unit-16
Week-8	Video-26	Formulas & Functions-I	Unit-17
	Video-27	Formulas & Functions-II	Unit-17
	Video-28	Formulas & Functions-III	Unit-17
	Video-29	Formulas & Functions-IV	Unit-17

	Video-30	Formulas & Functions-V	Unit-17
Week-9	Video-31	Graphical Presentation of Data Meaning and Intro	Unit-18
	Video-32	Graphical Presentation of Data using Spreadsheet	Unit-18
Week-10	Video-33	Advance Option in Spreadsheet-I	Unit-19
	Video-34	Advance Option in Spreadsheet-II	Unit-19
	Video-35	Advance Option in Spreadsheet-III	Unit-19
Week-11	Video-36	Google Workspace and Cloud	All Blocks (Unit 1-20)
	Video-37	Burgeoning & Contemporary Thought In business and Computers	All Blocks (Unit 1-20)
Week-12	Video-38	FAQ-I Computer application	All Blocks (Unit 1-20)
	Video-39	FAQ-II Business Application	All Blocks (Unit 1-20)

	Video-40	Terminology used in Computer Application in Business	All Blocks (Unit 1-20)
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Books and references

Issue Date	Title	Authors and Contributors
2020	<u>Unit-4 Basics of Computer Software</u>	<u>Dhusia, D.K.</u>
2020	<u>Unit-3 Web Applications</u>	<u>Dhusia, D.K.</u>
2020	<u>Unit-2 Application of Computers</u>	<u>Dubey, Arvind Kumar</u>
2020	<u>Unit-1 Introduction to Computer</u>	<u>Kesharwani, Subodh</u>
2020	<u>Block-1 Fundamentals of Computers</u>	<u>Kesharwani, Subodh; Dubey, Arvind Kumar; Dhusia, D.K.; Kesharwani, Subodh</u>

Criteria to get a certificate:

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

CTME102 Basics of Digital Marketing

Description:

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The course familiarize students with an understanding of how the digital media works and develop the critical insights necessary to succeed in e-commerce and digital and social media marketing. Enrich learner with the essential principles and practices of marketing in the digital economy by making this course, an indispensable part of their knowledge base. Companies that were non-existent a few short years ago, including Facebook, Whatsapp, LinkedIn, and Youtube, have dramatically changed how we interact, communicate and get from point A to B. Even Amazon and Google are relatively “young” by any measure. This course is also to serve as basic course for learners who wish to further study in the domain of Digital Media.

Pre-requisites:

Person who wish to study in the domain of Digital Media

Course layout:

Week 1	W1_L1 Introduction to Marketing W1_L2 Consumer Behavior W1_L3 Introduction to Digital Marketing
Week 2	W2_L1 Search Engine Optimization 1 W2_L2 Search Engine Optimization 2 W2_L3 Search Engine Optimization 3
Week 3	W3_L1 Search_Engine_Marketing : Part 1 W3_L2 Search_Engine_Marketing : Part 2 W3_L3 Display Advertising
Week 4	W4_L1 Social Media Marketing W4_L2 Social Media Marketing (Word of Mouth) W4_L3 Social Media Platforms Part 1 W4_L4 Social Media Platforms Part 2
Week 5	W5_L1 Online Reputation Management W5_L2 Mobile Marketing Part 1 W5_L3 Mobile Marketing Part 2

	W5_L4 Website Planning & Creation
Week 6	W6_L1 Email Marketing Part 1 W6_L2 Email Marketing Part 2 W6_L3 Introduction to E Commerce Part 1 W6_L4 Introduction to E Commerce Part 2
Week 7	W7_L1 E-COMMERCE Strategy 1 W7_L2 E-COMMERCE Strategy 2 W7_L3 Content Marketing W7_L4 Content Creation Process W7_L5 Influencer Marketing
Week 8	W8_L1 Digital Marketing Strategy Part 1 W8_L2 Digital Marketing Strategy Part 2a W8_L3 Digital Marketing Strategy Part 2b W8_L4 Digital Marketing Strategy Part 3a W8_L5 Digital Marketing Strategy Part 3b
Week 9	W9_L1 Digital Analytics & Measurement Part - 1a W9_L2 Digital Analytics & Measurement Part - 1b W9_L3 Digital Analytics & Measurement Part - 2a W9_L4 Digital Analytics & Measurement Part - 2b
Week 10	W10_L1 Digital Analytics & Measurement Part - 3a W10_L2 Digital Analytics & Measurement Part - 3b W10_L3 Measuring Campaign Effectiveness : ROI – Return on Investment W10_L4 Measuring Campaign Effectiveness : CLV - Customer Lifetime Value W10_L5 Attribution Models
Week 11	W11_L1 Digital Marketing Plan W11_L2 Case Study 1 : Email Marketing W11_L3 Case study 2 : Golf Tripz W11_L4 Case Study 3 : Golkonda Hotels W11_L5 Case Study 4 : Bhalaje Photography
Week 12	W12_L1 New Technologies & Advancement in Digital Marketing Part 1 W12_L2 New Technologies & Advancement in Digital Marketing Part 2 W12_L3 Review of Digital Marketing Course Part 1 W12_L4 Review of Digital Marketing Course Part 2

Books and references

1. Digital Marketing: Cases from India by Rajendra Nargundkar and Romi Sainy, Notion Press, Inc
2. Understanding Digital Marketing: Marketing Strategies for Engaging the Digital Generation by Damian Ryan, Kogan Page Publisher
3. Marketing 4.0: Moving from Traditional to Digital by Philip Kotler, Publisher Wiley
4. Digital Marketing by Seema Gupta, McGraw Hill Education
5. Fundamentals of Digital Marketing by Punit Singh Bhatia, Pearson
6. The Art of Digital Marketing: The Definitive Guide to Creating Strategic, Targeted, and Measurable Online Campaigns by Ian Dodson, Wiley Publisher

Criteria to get a certificate:

If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

CTUD101 - Fundamentals of Digital Electronics

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: In order to achieve the course objectives, students will be introduced to digital technologies. Various digital modules used to create digital computer devices like gates, flip flops, decoder, encoder etc. are to be studied. Mathematical base is created to understand the organization of digital computer.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Digital Systems and Number Systems	12
2	Boolean Algebra and Logic Gates	12
3	Combinational Circuit Designing	10
4	Sequential Circuit Designing	10
5	Memory Interfaces and Data Representation	08
6	The Arithmetic Logic Unit and Control Unit	08

Total Hours (Theory): 60

Total Contact Hours: 60

Detail Syllabus:

Unit I: Digital Systems and Number Systems

Hours 12

Introduction and Evolution of Digital computers, Major components of digital computer, Interfaces and Buses, compiler, interpreter, Assembler, Introduction to various number systems (Decimal, Binary, Octal, Hexadecimal, BCD), Conversion of number system, Binary operations, signed binary numbers, 1's complement and 2's complement.

Unit II: Boolean Algebra and Logic Gates

Hours 12

Fundamental concepts and rules of Boolean algebra, , Boolean Algebra Laws, Boolean expressions and its simplification with and without Truth Tables, Canonical form and standard form, D-Morgan's Theorem, Basic duality laws, Derivation of Boolean expressions, Introduction to logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), Implementation of Boolean expressions using logic gates, Sum of product and Product of Sum forms, Universal gates, Implementation of other gates using universal gates, Karnaugh Map method for two, three, four and five variables, K-map with don't care conditions, Tabulation method, determination and selection of prime applicants.

Unit III: Combinational Circuit Designing

Hours 10

Introduction to combinational circuits, Design procedures, Adders, Subtractors, code conversion circuits, Analysis procedures, Multilevel NAND circuits, Multilevel NOR circuits, EX-OR functions, Binary adder and Subtractors, Decimal adder and Subtractors, Magnitude comparator, Decoder and encoders, Multiplexer and de-multiplexer.

Unit IV: Sequential Circuit Designing

Hours 10

Introduction to Sequential circuits, Concepts of Flip-flops, triggering flip-flops, Clock signals, Analysis of clock sequential circuits, State Reduction and assignments, Flip flop excitation table, Design procedure of asynchronous and synchronous counters (Ripple counter, Binary counter, BCD counter), Design of registers, Shift registers, Timing sequences.

Unit V: Memory Interfaces and Data Representation

Hours 08

Memory, Types of ROM, Design of ROM, Types of RAM, Magnetic disc memory, Magnetic tape, Digital recording techniques, Linear-select memory organization, Types of digital codes (Gray code, 8421 code, Alphanumeric codes), Parity checking codes, Floating point representation, Fixed point representation.

Unit VI: The Arithmetic Logic Unit and Control Unit

Hours 08

Basic operations of Arithmetic Logic Unit, Construction of the Arithmetic Logic Unit, Representation of Instruction word, Control registers, Instruction and execution cycles of Control registers, Sequence operation of Control Registers.

Core Books:

1. M. Morris Mano: Digital Logic and Computer Design, Third Edition, Pearson Education, 2016.
2. Thomas C. Bartee: Digital Computer Fundamentals, Sixth Edition, Tata McGraw Hill Publishing, 2012.
3. M. Morris Mano: Computer System Architecture, Third Edition, Pearson Education, 2011.

Reference Books:

1. Andrew S. Tanenbaum: Structured Computer Organization, Fourth Edition, Pearson Education, 2005.
2. Albert Paul Malvino and Jerald A. Brown: Digital Computer Electronics, Third Edition, Tata McGraw Hill Publishing, 2008.

Web References:

1. https://www.tutorialspoint.com/digital_circuits/index.htm [Digital Circuits Tutorial]
2. <https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/> [Digital Electronics and Logic Design Tutorials]
3. <https://studymaterialz.in/digital-logic-and-computer-design-by-morris-mano/> [Digital Logic and Computer Design book]
4. <https://cupola.gettysburg.edu/oer/1/> [Digital Circuit Projects]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to get the idea about digital system and numbering system.
CO2 :	Able to study logic gates for digital circuit designing.
CO3 :	Able to learn to design various combinational circuits.
CO4 :	Able to learn to design various sequential circuits.
CO5 :	Able to learn memory design concepts and to study computer Arithmetic Logic Unit and Control Unit.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Digital Systems and Number Systems	√				
2	Boolean Algebra and Logic Gates		√			
3	Combinational Circuit Designing			√		
4	Sequential Circuit Designing				√	
5	Memory Interfaces and Data Representation					√
6	The Arithmetic Logic Unit and Control Unit					√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	2	2	-	3	2	-	-	2	1	3	2
C02	3	2	2	2	-	3	2	-	-	2	1	3	2
C03	3	3	3	2	-	2	2	-	-	2	2	3	2
C04	3	3	3	2	-	2	2	-	-	2	2	3	2
C05	3	3	3	2	-	3	2	-	-	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUS101 - Mathematics for Computer Science

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Theory	02	30	25	55
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: Basic algebra and geometry, as well as a good understanding of logic and proofs, mathematical statistics.

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of mathematics and statistics. Problem solving: Students will demonstrate the ability to solve problems, including applications outside of mathematics. Mathematical communication: Students will demonstrate the ability to communicate mathematical ideas clearly. They will use correct mathematical terminology and proper mathematical notation.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Sets and functions	04
2	Vectors and Matrices	04
3	Fundamentals of Calculus	06
4	Permutation & Combination	05
5	Probability	06
6	Graph Theory	05

Total Hours (Theory): 30
Total Contact Hours:30

Detail Syllabus:

Unit I : Sets and Functions

Hours 04

Introduction to set theory, methods of representation of a set, operations on set, algebra of sets, De 'Morgan's law, function definition, domain, range, one-to-one function, onto function.

Unit II : Vectors and Matrices

Hours 04

Definition of vector, addition and subtraction of vector, magnitude of a vector, unit vectors, dot product and cross product, definition of matrix, equal matrices, diagonal elements of matrix, row matrix, column matrix, Symmetric matrix, skew symmetric matrix, orthogonal matrix, diagonal matrix, identity matrix, operations on matrix.

Unit III : Fundamentals of Calculus

Hours 06

Limits, Continuity and Differentiability, Successive Differentiation, Partial Differentiation, Tangents and Normals, Maxima and Minima.

Unit IV : Permutation and Combination

Hours 05

Meaning of permutation, Formula of permutation, Permutation of n different things, Permutation of similar things, Permutation of repeated things, Circular Permutation, Combination: Meaning of Combination, Formula of Combination.

Unit V : Probability

Hours 06

Probability: Random Experiment, Sample Space, Event, Mutually exclusive event, Exhaustive event, Equally likely event, Probability Classical definition. (Simple examples of Probability).

Unit VI : Graph Theory

Hours 05

Introduction to Graph, Graph Definition, Vertices, Edges, Loops, Parallel Edges, Simple Graph, Finite Graph, Adjacent vertices, Incidence between vertex and edge, Degree of a vertex, Isolated Vertex, Pendent Vertex, Null Graph. Isomorphism, Labeled Graph

Core Books:

1. D. C. Sancheti, V. K. Kapoor: Business Mathematics, Sultan Chand & sons.
2. Lipschutz & Marc Lipson: DISCRETE MATHEMATICS, Tata McGraw Hill
3. Narsingh Deo: Graph Theory with application to engineering and computer science, Prentice Hall of India Pvt. Ltd

Reference Books:

1. Gupta and Gupta: Business Statistics, Sultan Chand and Sons.

Web References:

1. <https://www.statlect.com/matrix-algebra/vectors-and-matrices>
2. <https://www.geeksforgeeks.org/permutations-and-combinations/>
3. <https://www.khanacademy.org/math/statistics-probability/probability-library/basic-theoretical-probability/a/probability-the-basics>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will learn basics of sets and functions
CO2 :	Students will be familiar with various linear algebra.
CO3 :	Students will be able to compute limits, derivatives, and integrals. Analyze functions using limits, derivatives, and integrals
CO4 :	Students will be able to define a permutation and explain how to calculate probability
CO5 :	Students will be able to understand and apply the fundamental concepts in graph theory

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Sets and Functions	√				
2	Vectors and Matrices		√			
3	Fundamentals of Calculus			√		
4	Permutation and Combination				√	
5	Probability				√	
6	Graph Theory					√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	3	3	2	-	2	1	-	-	-	-	2	-
C02	3	3	3	2	-	2	1	-	-	-	-	2	-
C03	3	3	3	2	-	2	1	-	-	-	-	2	-
C04	3	3	3	2	-	2	1	-	-	-	-	2	-
C05	3	3	3	2	-	2	1	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**TEACHING SCHEME &
DETAILED SYLLABUS**

FOR

**B.Sc. (IT) PROGRAMME
(2nd SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2024-25**

B.Sc. (IT) Semester-II

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC103	Introduction to Object Oriented Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC104	Fundamentals of Web Designing	Practical	-	-	25	25	02	Major Core
CTUE103 CTUE104	(Select any one) 3. Business Models for e-Commerce 4. Managerial Economics	Theory	50	50	-	-	04	Minor
CTUD102	Data Communication and Networks	Theory	50	50	-	-	04	MDC
CTUV101	Fundamentals of Accounting	Theory	25	25	-	-	02	VAC
HSUS101-117	(Select any one) A course on Liberal Arts HSUS101 – Painting HSUS102– Photography HSUS103– Sculpting HSUS104– Pottery and Ceramic Art HSUS105-Media and Graphic Design HSUS106-Art and Craft HSUS109- Dramatics HSUS110-Contemporary Dance HSUS111-Music (Vocal) HSUS112-Music (Instrumental)-Tabla HSUS113-Music (Instrumental)-Guitar HSUS114-Music (Instrumental)-Harmonium HSUS115-Music (Instrumental)- Flute HSUS116- Indian Classical Dance-Kathak HSUS116- Indian Classical Dance-Bharatnatyam	Practical	-	-	25	25	02	SEC
CTUA101	Presentation Skills	Practical	-	-	25	25	02	AEC
CTUI101	Summer Internship and Viva - I	VIVA	-	-	50	50	4	Exit Course
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

CTUC103 - Introduction to Object Oriented Programming

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory) : Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical) : Lab Assignments, Practical Test, VIVA					

Pre-requisite: Introduction to Programming

Methodology, Pedagogy & Andragogy: The theory sessions will be enriched with real-world examples to demonstrate how object-oriented programming (OOP) enhances problem-solving. These examples will showcase the practical application of OOP concepts such as classes, objects, inheritance, polymorphism, and exception handling. By applying these concepts to real-world scenarios, students will gain a deeper understanding of how OOP can be used to solve complex problems efficiently and effectively. During the practical sessions of the Practical Programming Subject course, students will apply the concepts through a variety of assignments and case studies to assess their comprehension and progress.

Outline of the course(Theory):

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Object Oriented Programming	08
2	Classes and Objects	11
3	Inheritance	13
4	Polymorphism	10
5	Exception Handling and Type Cast Operators	10
6	Managing I/O formats and operations	08

Total Hours (Theory): 60

Total hours (Practical): 60

Total Contact Hours: 120

Detailed Syllabus:

Unit - I: Introduction to Object Oriented Programming

Hours: 08

Structured programming vs. object oriented programming, Introduction to C++: structure of a C++ program Tokens, inline function, pass by reference, default arguments, OOP characteristics.

Unit - II: Classes and Objects

Hours: 11

Define class, objects, visibility modes, static members, friend function, Constructors and Destructors, Default Constructor, Copy constructors, Parameterized Constructor, this keyword.

Unit - III: Inheritance

Hours: 13

Objective of Inheritance, types of inheritance, impact of access modes during inheritance, Virtual base classes, Dynamic Binding, Virtual functions, Pure virtual functions, Abstract class, Early vs. Late binding.

Unit - IV: Polymorphism

Hours: 10

Introduction of Static Binding, Function Overloading and ambiguity raised in function overloading, Operator overloading through member function and friend functions.

Unit - V: Exception Handling and Type Cast Operators

Hours: 10

Introduction to Exception handling, catching class types, using multiple catch statements, exception handling options, creating custom exceptions, introduction to cast operators, const cast, static cast, reinterpret cast, dynamic cast.

Unit - VI: Managing I/O Formats and Operations

Hours: 08

Understand and apply concepts from various header files like `iostream`, `iomanip`, `fstream`, etc. opening and closing file, reading and writing text files and binary files, performing random access on files, handle command line arguments.

Outline of the course(Practical):

Week No	Practical	Description
1	Introduction to Object Oriented Programming and Setting up IDE	Introduction to C++, Setting up a development environment and its introduction, structure of a C++ program Tokens, inline function, pass by reference, default arguments
2	Classes and Objects	Define class, objects, visibility modes, static members
3	Friend Functions	Introduction and hands on friend functions
4	Constructors and Destructors	Constructors and Destructors, Default Constructor, Types of constructs
5	Inheritance	Exposure on Inheritance programs and their types
6	Virtual Function	Programs on Virtual Function, Pure Virtual functions, Abstract Class, Early Binding and Late Binding
7	Function Overloading	Introduction of Static Binding, Function Overloading and ambiguity raised in function overloading
8	Operator Overloading	Operator overloading through member function and friend functions
9	Exception Handling	Introduction to Exception handling, catching class types,
10	Multiple Catch Statements	Using multiple catch statements, exception handling options, creating custom exceptions
11	Type Cast Operators	Introduction to cast operators, const cast, static cast, reinterpret cast, dynamic cast
12	Managing I/O Formats	Understand and apply concepts from various header files like iostream, fstream, etc
13	File Handling	Opening and closing file, reading and writing text files and binary files
14	File Handling Operations	opening and closing file, reading and writing text files and binary files, performing random access on files
15	Command Line Arguments	handle command line arguments.

Total hours: 60**Core Books:**

1. Herbert Schildt: C++: A Beginner's Guide, 2nd Edition Paperback, McGraw-Hill, 2003.
2. E Balagurusamy: Object Oriented Programming with C++ Paperback, McGraw Hill India, 2017.

Reference Books:

1. Yashavant Kanetkar: Let Us C++, Paperback, BPB Publications, 2020.
2. Robert Lafore: Object Oriented Programming in C++, 4th Edition, Sams Publications, 2002.
3. Bruce Eckel: Thinking in C++, Volume 1, 2nd Edition, Pearson Education, 2006.

Web References:

1. <https://www.geeksforgeeks.org/c-plus-plus/> [For Entire Syllabus]
2. <https://google.github.io/styleguide/cppguide.html> [For coding standards and concepts]
3. http://www.tutorialspoint.com/cplusplus/cpp_basic_input_output.htm [For Practical's]

Course Outcomes: Upon successful completion of the course, the students will:

C01 :	Understand the basic structure of C++ program and understand the concept of object oriented programming characteristics.
C02 :	Understand the concept of Constructor and Destructor and able to create class and objects.
C03 :	Understand the need of Polymorphism and implement static polymorphism.
C04 :	Understand the purpose of Inheritance, identify relations between classes, implement inheritance and perform dynamic binding.
C05 :	Able to handle the runtime errors or exceptions in C++ and use type cast operators. Able to read input from device, file, command line arguments and apply formats to input and output.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Object Oriented Programming	√				
2	Classes and Objects		√			
3	Inheritance			√		
4	Polymorphism				√	
5	Exception Handling and Type cast operators					√
6	Managing I/O formats and operations					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	2	2	2	2	2	2	2	2	2	2	2	3	2
C02	3	3	3	2	2	2	3	2	2	2	2	3	2
C03	3	2	3	2	2	2	3	2	2	2	2	3	2
C04	3	3	3	2	2	2	3	2	2	2	2	3	2
C05	3	2	3	2	2	2	2	2	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUC104 - Fundamentals of Web Designing

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components : Lab Assignments, Practical Test, VIVA					

Prerequisites: Basic concepts of web.

Methodology, Pedagogy & Andragogy: This course focuses on providing hands-on experience to students for design and develop entire web sites using several web designing tools and HTML scripting language.

Outline of the Course:

Week No	Practical	Description
1	Basic Construction of an HTML & HTML5 Page, HTML Text Editors, HTML Building blocks.	Learn basic concepts of web page design.
2	Basic HTML tags – Text formatting tags, Heading tags, paragraph tags, division tag, Span tags, Break line tag, Binding space in HTML .	Get familiar with different tags and attributes which is used to add the text in a web page.
3	Controlling Font size and color, Adding Table – Table tag.	Learn user interface designing using Table Layout, change font style, color etc.
4	Grouping and merging table rows and columns. Adding List – ordered, unordered and definition list.	Able to display the data in various formats and design different types of list in the table.

5	Adding Image, Adding links - Changing link colors, Email link.	Learn to add images and link in the web page.
6	Embedding audio and video, Adding Field sets.	Able to add multimedia in web page using internal and external methods.
7	Adding Form: Form tag, Action, Method. Basic form controls.	Learn basic form elements and when and where to apply it.
8	Textbox, Password field, Radio Button, Checkbox, Multiple line text area, file upload.	Use various types basic form controls with types of form input elements.
9	Various types of buttons – submit, reset, image, simple button.	Learn various types of buttons to make it more interactive.
10	HTML5 input elements - Color, Number, Date, Month.	Learn new input types in HTML5.
11	Week, Time, date time- local, Email, url, range, search.	Learn to add defining inputs and minimum and maximum range to make the form attractive.
12	Data list, output, progress, meter.	Learn to add list and provide progress horizontal view and range with output.
13	Regular expression, HTML5 new attributes – Placeholder, Required.	Learn HTML 5 form element to provide user side web form validation.
14	Pattern, Autocomplete, Autofocus, novalidate, formnovalidate.	Learn HTML 5 new form properties.
15	formaction, formmethod, spellcheck, contenteditable.	Learn HTML 5 to user submit the form and redirect the form data to selective method.

Total hours: 60

Text Books:

1. Powel Thomas: HTML - The Complete Reference, 3rd Edition, McGraw-Hill Education – Europe, 2018.
2. DT Editorial Services: HTML 5 Black Book, 2nd Edition, Dreamtech Press, 2016
3. Matt West: HTML 5 Foundations, Wiley publication, 2013
4. Farrar: HTML Example book, BPB, 2007.

Reference Books:

1. David DuRocher : HTML and CSS QuickStart Guide, Paperback, 2021
2. Laura Lemay, Rafe Colburn, Jennifer Kyrnin : Mastering Html, CSS & JavaScript Web Publishing, Paperback, 2016.
3. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.
4. Shelly Woods: HTML introductory concepts and techniques, 5th Edition, Course Technology, 2009.
5. Jon Duket: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.

Web References:

1. www.w3schools.com/html [For HTML tutorials]
2. html.net/tutorials/html/ [For HTML tutorials]
3. www.htmlgoodies.com/ [For HTML resources]
4. <https://developer.mozilla.org/en-US/learn/html> [For HTML best practices]

Course Outcome: Upon successful completion of the course, student will:

C01 :	Able to understand basic construction of HTML & HTML5 document and started using basic tags.
C02 :	Able to Add content using various tags and format the content in table & list.
C03 :	Able to design links and multimedia elements such as image, audio, video.
C04 :	Able to design a web page form using HTML & HTML5.
C05 :	Able to apply validation using HTML5 attributes.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO1 1	PSO1	PSO2
C01	1	2	1	3	2	2	1	1	-	1	1	1	1
C02	-	1	2	3	1	3	1	-	-	2	2	2	2
C03	1	1	2	3	1	3	1	-	-	2	2	2	2
C04	1	1	2	3	1	1	1	1	-	1	-	1	1
C05	2	2	3	3	2	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUE103 - Business Models for e-Commerce

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of E-commerce. Students will be introduced to E-Commerce, E-Marketing, E-Security and E-Payment, Mobile Commerce, CRM and ERP, and E-Commerce Applications. Students will give practical exposure in form of case study and by visit to E-commerce / Digital Marketing company and/or supply chain Centre.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to E-Commerce	09
2	E-Marketing	11
3	E-Security and E-Payment	09
4	Mobile Commerce	11
5	E-CRM and ERP	09
6	E-Commerce Applications	11

Total Hours (Theory): 60

Total Contact Hours:60

Detail Syllabus:

Unit I : Introduction to E-Commerce

Hours 09

Emergence and use of Internet, origin of web and world wide web, advantages and disadvantages of E-commerce, Features of E-Commerce, Electronic Commerce over the Internet, Types of Ecommerce.

Unit II : E-Marketing

Hours 11

Traditional Marketing, Online Marketing, E- advertising, Internet Marketing Trends and strategies, E-branding.

Unit III : E-Security and E-Payment

Hours 09

Security on the Internet, E-Business risk management issues, Digital token based E-Payment System, Properties of Electronic Cash, Digital Signature.

Unit IV : Mobile Commerce

Hours 11

Definition of M-Commerce, Features of M-Commerce, Advantages and Disadvantages of M-Commerce, Areas of M-Commerce Applications, Payment Systems and Models in M-Commerce.

Unit V : E-CRM and ERP

Hours 09

Customer Relationship Management, CRM capabilities and Customer life cycle, Introduction to ERP, Reasons for the growth of the ERP Market, Advantages and Disadvantages of ERP.

Unit VI : E-Commerce Applications

Hours 11

Retail and Wholesale, Finance, Manufacturing, Online Booking, Online Publishing, Digital Advertising, Digital Shopping, Digital Media, Auctions, A case study need to be discussed for these application. Hands-on experience in setting up and managing an e-commerce store. Developing a business plan for an ecommerce startup.

Core Books:

1. P.T.Joseph, S.J.: E-Commerce- An Indian Perspective, 3rd Edition, PHI learning Private Limited.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education.

Reference Books:

1. Anita Rosen: The e-commerce Question and Answer Book: A Survival Guide for Business Managers, 2nd Edition, Amacom.
2. Janice Reynolds: The Complete E-Commerce Book: Design, Build & Maintain a Successful Web-based Business, 2nd Edition, CRC Press.
3. By Philippe Humeau&Matthieu Jung : The White Book of Ecommerce Solutions, NBS System.
4. IshitaLahiri and Sujit Kumar Ghose : Principals of Marketing and E-Commerce

Web References:

1. <http://www.ecommercetutorial.net/> [For Tutorial]

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	To introduce the concept of Electronic Commerce
C02 :	To understand the concept of E-marketing
C03 :	To understand the role of E-Security & E-Payment
C04 :	To understand M-Commerce Payment System
C05 :	To learn CRM, ERP and E-commerce Applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to E-Commerce	√				
2	E-Marketing	√	√			
3	E-Security and E-Payment		√	√		
4	Mobile Commerce		√	√		
5	E-CRM and ERP				√	
6	E-Commerce Applications					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	2	2	3	2	2	1	2	2	2	1	1	3	3
C02	3	3	3	3	3	3	3	3	3	3	3	3	3
C03	2	2	3	3	2	2	2	2	2	1	1	3	2
C04	3	3	3	3	3	3	3	3	3	2	3	3	2
C05	3	3	3	3	3	3	3	3	3	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUE104 - Managerial Economics

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theoretical lessons, real-world examples will be used to discuss economic topics such as demand, supply, production, cost function, market structure, factor pricing, and macroeconomics. Students will be offered appropriate case studies in order to provide actual exposure to theoretical economic ideas.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Basic Economic Concepts	08
2	Demand and Supply	11
3	Production, Cost and Revenue	11
4	Inflation	11
5	Market Structures & Pricing	10
6	Market Failures and Price Regulations	09

Total Hours (Theory): 60

Total Contact Hours: 60

Detail Syllabus:

Unit I: Basic Economic Concepts

Hours 08

Basic of Economics, Nature of Economics, Scope of Economics, Business Economics, Difference Between Micro and Macro Economics, Laws of Economics, Business Cycle, Inflation, Needs, wants.

Unit II: Demand and Supply

Hours 11

Demand and supply, Types of Demand, Determinants of Demand and supply, Law of Demand and supply Demand and supply Schedule, Demand Function. Own-price elasticity of demand, cross-price elasticity of demand, and income elasticity of demand.

Unit III: Production, Cost and Revenue

Hours 11

Production function, law of variable proportion and laws of returns to scale, different types of costs – variable cost, fixed cost, total cost, average cost, average fixed cost, average variable cost and marginal cost, Total revenue, average revenue and marginal revenue, profit function.

Unit IV: Inflation

Hours 11

Meaning, Depreciation of money, Monetary inflation concepts, types of inflation, Causes and effect of inflation different sectors of the economy, Causes of Inflation, Advantages, Formula and Disadvantages of inflation.

Unit V: Market Structures & Pricing

Hours 10

Concept of market and equilibrium- characteristics of perfect competition, Features of Market Structure, monopoly, monopolistic competition and oligopoly–price determinations, Measure of market structure with example.

Unit VI: Market Failures and Price Regulations

Hours 09

Market failures and need for regulation, Regulations and market structure, Firm behavior, Price regulation. More on Externalities, and role of government, importance of competition in market failure and price regulations.

Core Books:

1. P.L Mehta: Managerial Economics, Sultan Chand and Sons, (2014)
2. Keat Paul , K Young Philip , Erfle Steve , College Dickinson, Banerjee Sreejatha , Managerial Economics, Pearson Education; 6th ed..

Reference Books:

1. Yogesh Maheswari, Managerial Economics, Phi Learning, New Delhi, 2005 Gupta G.S.
2. Moyer & Harris. Managerial Economics, Tata Mcgraw-Hill, New Delhi
3. Geetika, Ghosh & Choudhury. Managerial Economics, Cengage Learning, New Delhi, 2005
4. Dominick Salvatore, Managerial Economics, adapted by Ravikesh Srivastava, Oxford University press
5. Mote V.L., Samuel Paul and G.S. Gupta, Managerial Economics Concepts and Cases, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2001

Web References:

1. <https://www.geektonight.com/managerial-economics-notes/>
2. <https://www.learningclassesonline.com/2020/10/pedagogy-of-economics.html>
3. <https://mrcet.com/downloads/MBA/Managerial%20Economics.pdf>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand basic economic concepts.
CO2 :	Able to understand market.
CO3 :	Able to learn basics of inflation.
CO4 :	Able to deal with situation between cost and revenue.
CO5 :	Able to understand basics of Demand and Supply.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basic Economic Concepts	√				
2	Demand and Supply					√
3	Production, Cost and Revenue				√	
4	Inflation			√		
5	Market Structures & Pricing		√			
6	Market Failures and Price Regulations		√			

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	-	-	-	-	1	2	1	-	-	-	1	1	2
C02	-	1	1	-	1	2	2	-	-	1	3	1	2
C03	-	1	1	-	1	2	2	-	-	-	2	1	2
C04	1	1	1	-	1	2	2	-	-	1	2	1	2
C05	1	1	1	-	1	2	2	-	-	1	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUD102 - Data Communications and Networks

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study, Certification courses, etc.					

Pre-requisite: Basic knowledge of computers and their working.

Methodology, Pedagogy & Andragogy: During the lecture sessions, the students will learn about role of computer network in modern communication systems, underlying principles of network communication, need for standardization through network reference model. The teacher will also discuss communication issues at different layers and concepts related to network security. The students will adopt the latest certification courses from reputed and recognized platforms. Furthermore, the students will be engaged in problem-solving, emphasizing real-world applications and relevance.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Computer Networks	10
2	Physical Layer Communication	08
3	Network reference model	11
4	Data Link Layer Functions and Protocol	12
5	Network Layer Functions and Protocol	10
6	Network security	09

Total Hours (Theory):60

Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to computer network

Hours 10

Definition of computer network, aim and objective of computer network: applications of computer network, merits & demerits of computer network, examples of computer network, components of computer network, types of networks, network topologies, network devices: hub, switch, repeater, router, gateway, bridge.

Unit II: Physical Layer Communication

Hours 08

Analog and digital signal, Definition of bandwidth, Maximum data rate of a channel, Line encoding schemes, Transmission modes, Modulation techniques, Multiplexing techniques- FDM and TDM, Transmission media-Guided and Unguided, Switching techniques- Circuit switching, Packet switching, Connectionless datagram switching, Connection-oriented virtual circuit

Unit III: Network reference model

Hours 11

Need for network reference model, history of TCP/IP and OSI model, layers of OSI and TCP/IP model, comparison between TCP/IP and OSI model, Protocol Hierarchy, Network Protocols: Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Multiple Internet Mail Extension (MIME), Post Office Protocol (POP), Telnet, Domain Name Service (DNS), Dynamic Host Configuration Protocol (DHCP), Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP).

Unit IV: Data Link Layer Functions and Protocol

Hours 12

Definition of Framing, Framing methods, Error detection techniques, Error correction techniques, Flow control mechanisms- Simplex protocol, Stop and Wait ARQ, Go-Back-N ARQ, Point to Point protocol.

Unit V: Network Layer Functions and Protocol

Hours 10

Connection oriented vs Connectionless services, Definition of Routing, Routing algorithms, IP protocol, IP addresses, ARP, RARP

Unit VI: Network security

Hours 09

Introduction to cryptography, symmetric key and public key encryption systems, digital signature and key management concepts, communication security.

Core Books:

1. Andrew S. Tanenbaum: Computer Networks, 6th edition, Pearson Publication: 2021.
2. Behrouz Forouzan: Data communications and networking, 4th Edition, McGraw Hill, 2018.
3. William Stallings: Data and computer communication: 5th Edition, Pearson Publication, 2019.

Reference Books:

1. James F. Kurose, Keith W. Ross: Computer networking – A top-down approach: 3rd Edition: Pearson Publication: 2017
2. Narsimha Karumanchi: Elements of computer networking: An Integrated Approach: 1st Edition: Careermonk Publication: 2014
3. Ed Tittel: Computer networking: 1st Edition, Schum's Publication: 2020

Web References:

1. <https://computer.howstuffworks.com/computer-networking-channel.htm> [How Stuff Works]
2. [https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2012-R2-and-2012/dn313100\(v=ws.11\)?redirectedfrom=MSDN](https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2012-R2-and-2012/dn313100(v=ws.11)?redirectedfrom=MSDN) [Microsoft Networking]
3. https://help.sap.com/docs/SAP_COMMERCE/d0224eca81e249cb821f2cdf45a82ace/8c74877b866910148cc9ba5f39a2fd28.html?version=6.7.0.0 [For multicast and unicast]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Student will be able to understand role of computer network in digital communication
CO2 :	Student will grasp underlying concepts related with network communication
CO3 :	Student will understand role of reference models in standardizing communication
CO4 :	Students will learn about intra and inter network communication and technology behind it.
CO5 :	Students will learn about different aspects of network security.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Computer Networks	√				
2	Physical Layer Communication		√			
3	Network reference model			√		
4	Data Link Layer Functions and Protocol				√	
5	Network Layer Functions and Protocol				√	
6	Network security					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	2	3	2	1	-	1	1	1	-	-	1	2	-
C02	2	3	3	2	-	1	1	1	-	-	1	3	1
C03	3	3	3	2	-	1	1	1	1	-	1	3	3
C04	1	3	3	2	-	1	1	1	1	-	1	3	2
C05	3	3	3	2	-	1	1	1	1	-	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUV101 - Fundamentals of Accounting

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Theory	02	30	20	50
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During the classroom sessions, the emphasis will be given on the fundamentals of accounting. Students will be introduced to book keeping systems in modern accounting, financial accounting, and cost accounting. Students will prepare financial statements such as journal, ledger, trail balance, trading and profit and loss account, and balance sheet. Students will also get hands on exposure in make or buy decisions in cost accounting. Same concepts will be made concrete through assignments, case studies, and unit tests.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Accounting	05
2	Book Keeping System	05
3	Financial Accounting - I	05
4	Financial Accounting - II	07
5	Cost Accounting - I	04
6	Cost Accounting - II	04

Total Hours (Theory): 30

Total Contact Hours:30

Detail Syllabus:

Unit I : Introduction to Accounting

Hours 05

Introduction and users of Accounting, Branches of Accounting, Objectives, Accounting Concepts and Conventions, Accounting Terms

Unit II : Book Keeping System

Hours 05

Single and Double Entry Book Keeping Systems, Advantages of Double Entry Book Keeping System, Limitations of Double Entry Book Keeping System, Types of Accounts: Personal, Real, and Nominal, Terms of Book Keeping Systems

Unit III : Financial Accounting – I

Hours 05

Stages of Accounting Cycle, Types of Transactions, Forms of Economic Transactions, Cash and Credit Transactions, Introduction to Journal, Journal Entries

Unit IV : Financial Accounting – II

Hours 07

Introduction to Ledger, Relationship Between Journal and Ledger, Rules of Ledger Posting, Balancing of the Accounts in Ledger, Trial Balance, Objectives of Preparing Trial Balance, Trading and Profit and Loss Account, Balance Sheet, Difference between Trial Balance and Balance Sheet

Unit V : Cost Accounting – I

Hours 04

Introduction to Cost Accounting: Cost, Costing, and Cost Accounting, Limitations of Financial Accounting, Difference Between Cost Accounting and Financial Accounting, Objectives of Cost Accounting,

Unit VI : Cost Accounting – II

Hours 04

Elements of Cost: Material Cost, Labor Cost, and Expenses, Techniques of Costing, Marginal Costing, Make or Buy Decision, Breakeven Analysis

Core Books:

1. Dr. S. N. Maheshwari: Principles of Management Accounting, Sultan Chand Publication. Eighteenth Edition, 2021
2. M. Pande: Financial Management, Vikas Publishing House, 2015
3. M.N. Arora: Cost Accounting, Vikas Publishing House, 2013

Reference Books:

1. Dr. S.N. Maheshwari: Advanced Accountancy, Volume 1, Eleventh & Revised Edition, Vikas Publishing House, 2017

Web References:

- <https://www.accountingtools.com/articles/basics-of-accounting.html> [For Basics of Accounting]
- https://www.youtube.com/watch?v=vuetn_PQOvM [For Journal Entries – 1]
- <https://www.youtube.com/watch?v=3xCzh3-bm4o&list=RDCMUCNh1egMomGI3hjI0DHExdUg&index=3> [For Journal Entries – 2]
- https://www.youtube.com/watch?v=z_KO49Pk3DM [For Ledger Posting & Trial Balance]
- <https://www.youtube.com/watch?v=Y4azRCTTWoU> [For Trading A/c & Profit & Loss A/c]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will get basic understanding of the field of Accounting
CO2 :	Students will be familiar with Accounting Book Keeping Systems
CO3 :	Students will learn concepts of financial accounting and different types of accounts with rules of debit and credit
CO4 :	Students will be able to generate Journal, Ledger, Trial Balance, and Final Accounts
CO5 :	Students will get basic understanding of the field of Cost Accounting

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Accounting	√				
2	Book Keeping System		√			
3	Financial Accounting – I			√		
4	Financial Accounting – II				√	
5	Cost Accounting – I					√
6	Cost Accounting – II					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	1	1	-	-	2	2	3	1	2	1	1	1	1
C02	1	2	-	-	2	2	3	1	2	1	1	-	-
C03	3	3	2	1	1	2	3	-	2	2	1	2	-
C04	3	3	3	1	1	2	3	-	2	2	1	2	-
C05	2	3	2	1	3	2	3	1	2	3	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUA101 - Presentation Skills

Credit	Component	Instruction / Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	02	30	20	50
Experiential Learning Components: Lab Assignments, Practical Test, VIVA					

Prerequisites: Basic computer skills and knowledge of English.

Methodology, Pedagogy & Andragogy: The aim of this course is to develop effective presentation skills in students. Students will be taken systematically through the key stages of giving presentations, from planning and introducing to concluding and handling questions. The topics for the presentation will range from academic presentations to research presentations.

B. Outline of the Course:

Week No	Practical	Description
1	Introduction	Knowing the Purpose, Knowing the Audience, Internal and External Presentation, Presentation to Heterogenic Group, Greetings, Introducing self and peers, Asking and sharing information
2	ICT tools for presentation – Brief introduction	Microsoft PowerPoint, Canva, Prezi, AI-powered presentation tools
3	Presentation using Canva	Introduction to Canva, Creating, presenting, sharing and downloading presentation
4	Presentation design in Canva	Templates, layouts, Presenter Notes, Duration, Timer
5	Basic design elements in Canva	Shapes, Graphics, Charts, Tables
6	Advanced design elements in Canva	Photos, Videos, Audio, Frames, Grids, Stickers
7	Intermediate features of PowerPoint	Using Screenshot, Inserting objects, video and audio in presentation, Screen recording

8	Advanced features of PowerPoint	Using Hyperlink and Action, Slide Show Set-up, Presenter View, Master views
9	Cloud-based presentation using Google Slides	Creating a new presentation, importing slides, Sharing, Printing, Downloading in different formats, Version History
10	Basic features of Google Slides	Slideshow, Motion, Theme Builder, Inserting image, audio, video, shape
11	Advanced features of Google Slides	Spell-check, Linked objects, Q-A History, Notification Settings, Accessibility, Activity Dashboard
12	Engaging audience using feedback-gathering tools	Use of menitimeter for Word Clouds, Live Polls, Scales, Ranking and Pin It
13	Ethical practices	Using Citations/References for texts, images and other type of information, Plagiarism check
14	AI and Presentation	Use of Generative AI tools such as ChatGPT, CoPilot etc. in presentations
15	Select case-study	Examples of best presentations.

Total hours: 30

Text Books:

1. The Presentation Book : How to Create It, Shape It and Deliver It! Improve Your Presentation Skills Now, 2nd Edition, Pearson Education.
2. Beginners Guide for Canva 2024, by Karla J Kane. ISBN: 979-8875801853

Reference Books:

1. Microsoft PowerPoint Guide for Success by Kevin Pitch, Top Notch International. ISBN:978-1915331489
2. Getting Started with Google Slides: A Practical Guide to Cloud-Based Presentations by Scott La Counte, SL Editions ISBN:978-1629179513

Web References:

1. <https://www.canva.com/education/students/>
2. <https://support.microsoft.com/en-us/powerpoint>
3. <https://support.google.com/a/users/answer/9282488>
4. <https://www.mentimeter.com/features>

Course Outcome: Upon successful completion of the course, student will be able to:

C01	Understand key elements of an effective presentation and ICT tools for presentation.
C02	Acquire in-depth knowledge of one of the popular presentation tools.
C03	Develop intermediate and advanced skills in alternative presentation tool.
C04	Familiarize with cloud-based presentation tool.
C05	Create effective presentation with ethical practices and enhance it with AI tools.

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	2	1	3	-	2	1	-	-	2	-	-
C02	2	-	3	-	2	1	-	-	2	2	3	1	1
C03	-	3	-	2	-	1	3	2	-	-	-	3	2
C04	2	1	2	2	2	1	2	-	2	-	2	3	2
C05	-	2	2	3	1	-	2	2	2	2	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUI101 : Summer Internship and Viva – I

Objective and Scope:

Students who choose to exit the programme after the completion of the first two semesters must undertake internships, community engagement programmes and field based learning/minor projects during the summer term. This internship, worth 4 credits, is a requirement for the award of the Undergraduate Certificate. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. This provision ensures that students gain valuable practical experience and industry exposure, complementing their academic learning and enhancing their career prospects. Following are the intended objectives of internship training:

- **Practical Experience:** Provide hands-on experience with real-world projects to apply theoretical knowledge gained during the course.
- **Skill Development:** Enhance technical skills, such as programming, software development, database management, and other relevant IT skills.
- **Industry Exposure:** Familiarize students with the working environment of the IT industry, including understanding company culture, workflows, and professional practices.
- **Professionalism:** Instil a sense of professionalism and work ethic, including adhering to deadlines, following instructions, and working collaboratively.
- **Transition to Employment:** Facilitate a smoother transition from academic life to professional careers by bridging the gap between theory and practice.

Guidelines of Internship:

The Summer Internship shall be of 15 (Minimum 60 hours) duration and will be undertaken during the Summer Vacation. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. It is mandatory for the students to seek written approval from the coordinator about the topic and the organization before commencing the Summer Internship. During Summer Internship students are expected to take necessary guidance from the faculty guide allotted by the Institute. To do it effectively they should be in touch with their guide through e-mail or phone. Students must maintain regular reports detailing the work completed, challenges faced, and learning experiences. Students must submit a final report and presentation summarizing the internship experience, project outcomes, and key learnings to guide. The technical specifications for report preparation is as under:

Technical details:

1. The report shall be printed on A-4 size white paper.
2. 12 pt. Times New Roman font shall be used with 1.5-line spacing for typing the report.
3. 1" margin shall be left from all the sides.
4. Considering the environmental issues, students are encouraged to print on both sides of the paper.
5. The report shall be spiral bound as per the standard format of the cover page given by the Institute.
6. The report should include a Certificate (on company's letter head) from the company duly signed by the competent authority with the stamp.
7. The report shall be signed by the respective guide(s) & the Principal of the Institute 10 (Ten) days before the viva-voce examinations.
8. Student should prepare two hard bound copies of the Summer Internship Project Report and submit one copy in the institute. The other copy of the report is to be kept by the student for their record and future references.

Evaluation:

The evaluation of Internship will be done as per the following criteria:

Sr. No.	Evaluation Criteria	Marks
1	Summer Internship Report	50
2	Presentation and Viva-voce examination	50
TOTAL MARKS		100

Students must secure 36% passing marks in both components to qualify for the Certificate.

Internship and Exit:

Any student intending to exit the course must complete the summer internship (Including scoring at least 36% marks in the evaluation) before exit. A student intending to exit shall have to submit an application to Principal through Counsellor before even semester university theory examination. However, any such student desiring to withdraw the exit option may do that before the last paper of theoretical examinations of even semester.

**TEACHING SCHEME &
DETAILED SYLLABUS**

FOR

**BSc-IT PROGRAMME
(3rd SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2024-25**

BSc-IT Semester-III

Course Code	Course Name	Theory / Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC201	Introduction to Data Structures	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC202	System Analysis and Design	Theory	50	50	-	-	04	Major Core
CTUC203	Programming the Internet	Practical	-	-	25	25	02	Major Core
CTUS201	(Select any one) 3. Computer Oriented Numerical Methods	Theory	50	50	-	-	04	SEC
CTUS202	4. Computer Oriented Management System							
CTUA201	Life Management	Theory	25	25	-	-	02	AEC
	University Elective-I	Practical	-	-	50	50	02	MDC
HSUV201	Creativity, Problem Solving and Innovation	Practical	-	-	25	25	02	VAC

*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)
 *SEE- Semester –End- Evaluation

University Elective - I			
Sr. No.	Course Code	Course Name	Department/Faculty
1		Art of Programming	Engineering
2		Environmental Sustainability & Climate Change	Engineering
3		ICT Resources & Multimedia	Engineering
4		Engineering Drawing	Engineering
5		Fundamentals of Packaging	Pharmacy
6		Basic Laboratory Techniques	Applied Science
7		First Aid & Life Support	Nursing
8		Health Promotion & Fitness	Physiotherapy
9		Banking & Insurance	Management
10		Introduction to MATLAB Programming	Engineering
11		Astrophysics, Space and Cosmos-1	Applied Science
12		SDG Handprint Laboratory	Engineering
13		Python Programming	Engineering
14	CAUD201	Introduction to Web Designing	Computer Science

CTUC201: Fundamentals of Data Structures and Algorithms

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory) : Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical) : Lab Assignments, Practical Test, VIVA					

Pre-requisite: Foundation of C Programming, Fundamentals of Object Oriented Programming

Methodology, Pedagogy & Andragogy: Theory lectures focus on fundamental concepts of data structures and algorithms, emphasizing the selection of the most efficient structures for various applications. Andragogical principles are integrated to promote self-directed and practical learning for learners. For practical teaching of Data Structures and Algorithms, the approach combines hands-on programming exercises with real-world case studies to reinforce the selection of efficient structures for various applications. This blended method enhances both theoretical understanding and practical skills, ensuring learners can effectively apply concepts in real-world scenarios.

Outline of the course (Theory):

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1.	Principles of Data Structures and Algorithms	06
2.	Arrays and Sequential Data Structures	10
3.	Advanced Dynamic Data Structures	12
4.	Tree Structures and Their Applications	11
5.	Graph Theory and Utilizations	11
6.	Sorting and Searching	10

Total Hours (Theory): 60
Total hours (Practical) : 60
Total Contact Hours : 120

Detail Syllabus:

Unit I: Principles of Data Structures and Algorithms

Hours 06

Atomic and composite data, Data type, Data object, Data Structure, Abstract Data type, Types of Data Structures, Introduction to Algorithms, Relationship among Data, Data Structures and Algorithms, Analysis of Algorithms, space and time complexity algorithm

Unit II: Arrays and Sequential Data Structures

Hours 10

Linear Data Structures using sequential organization, Representation of Stacks using sequential organization, Applications of Stack, Application of Recursion, Concepts of Queues, Realization of Queues using sequential organization, circular queue, Multiqueue, Deque, Priority Queue, Applications of Queue.

Unit III: Advanced Dynamic Data Structures

Hours 12

Linked list, Comparison of sequential and linked organizations, Dynamic Memory Management, types of link list , Applications of linked list, operation on link list.

Unit IV: Tree Structures and Their Applications

Hours 11

Basic Terminologies, Definition and concepts, Representation of Binary Tree, Operations on Binary Tree and algorithms, Types of Binary Trees , BTree, B+Tree, ,Red and Black Tree, AVL trees,

Unit V: Graph Theory and Utilizations

Hours 11

Graph Terminology, Representation of Graph, Operations on Graph, Applications of Graph Structure, minimum spanning tree.

Unit VI: Sorting and Searching

Hours 10

Sorting Notations and Concepts, Sorting Techniques, Sequential Searching, Binary Searching, Search Trees.

Outline of the Course (Practical):

Week No	Practical	Description
1	Basic of C and C++	foundational concepts in C and C++.
2	Two dimension array	Understand and utilize arrays effectively.
3	Stack	Implement stack operations including push, pop, and peek.
4	Stack and its application	Cover infix, prefix, and postfix notations.
5	Queue and Circular queue	Implement queue operations such as insert, delete, and display.
6	Linear and Binary Search	Apply various searching techniques on arrays.
7	Merge sort, Bubble Sort	Implement and compare different sorting techniques to sort data.
8	Selection sort , Insertion Sort,	
9	Quick sort	
10	Single Link List	Perform insert and delete operations at the beginning, end, and middle of link list data structures.
11		
12	Double Link List	
13		
14	Circular Single Link List	
15	Circular Double Link List	

Total hours: 60**Core Books:**

1. Jean-Paul Tremblay, Paul G. Sorenson: An introduction to data structures with applications, 2nd Edition, Tata McGraw Hill Publications, 1991.
2. D. Samanta: Classic Data Structures, 2 Edition, PHI Publications, 2002.

Reference Books:

1. Varsha h. Patil : Data Structures Using C++, 1st Edition, Oxford University Press, 2012.
2. Yashvant Kanethkar : Data structures through C++, 2nd Edition, BPB Publications, 2003.
3. Mark Allen Weiss: Data Structures and Algorithm Analysis in C++, 3rd Edition, Pearson Education, 2009.
4. Reema Thareja: Data Structures using C, 1st Edition, Oxford University Press, 2012.

Web References:

1. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/ds_ToC.html [For materials of Data Structures and Algorithms]
2. <http://www.cs.usfca.edu/~galles/visualization/> [For Data Structures Visualization]
3. <https://www.cs.auckland.ac.nz/~jmor159/PLDS210/mst.html> [For Graph Data Structures]
4. <http://interactivepython.org/runestone/static/pythonds/Trees/trees.html> [For Tree Data Structures]
5. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/niemann/s_man.pdf [For Sorting and
i. Searching Cookbook]
6. <http://www.cs.princeton.edu/~rs/AlgsDS07/10Hashing.pdf> [For Hashing Materials]
7. <http://nptel.ac.in/video.php?subjectId=106102064> [For Data Structures videos]

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	Comprehend and convey the essential concepts and importance of data structures and algorithms.
C02 :	Effectively implement and use array-based and linear data structures.
C03 :	Analyze and apply advanced dynamic data structures, such as linked lists.
C04 :	Construct and manipulate various tree data structures, including binary trees and AVL trees.
C05 :	Use graph theory concepts to solve complex problems and understand real-world applications.
C06:	Implement and compare various sorting and searching algorithms for optimal performance.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Principles of Data Structures and Algorithms	√					
2	Arrays and Sequential Data Structures		√				
3	Advanced Dynamic Data Structures			√			
4	Tree Structures and Their Applications				√		
5	Graph Theory and Utilizations					√	
6	Sorting and Searching						√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	3	-	1	3	1	-	-	1	1	2	1
C02	3	2	3	1	1	3	1	-	-	1	2	2	1
C03	3	2	3	1	1	3	1	-	-	1	2	2	1
C04	3	2	3	1	1	3	1	-	-	1	2	2	1
C05	3	2	3	1	1	3	1	-	-	1	2	2	1
C06	3	2	3	1	1	3	1			1	3	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUC202 : System Analysis and Design

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding of analysing business needs, perform planning and analysis of business system, give examples of how Information Systems are designed and constructed and how major modifications are made to the existing systems.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Basics of Information Systems	08
2	Information Systems Analysis Overview	11
3	Information Gathering and System Requirement specification	11
4	Data flow diagrams and Process Specification	11
5	Overview to Data input and designing output methods	10
6	Implementation and maintenance of information systems	09

Total Hours (Theory): 60

Total Contact Hours:60

Detailed Syllabus:

Unit - I: Basics of Information Systems

Hours:08

Types of Information, Understand the concept of System, Need for a Computer-Based Information

System, Management Structure, Management and Information Requirements, Qualities of Information, Examples of Information Systems, Various Functions in Organizations, Varieties of Information Systems.

Unit - II: Information Systems Analysis Overview

Hours: 11

Overview of Design of an Information System, System Development Life Cycle (SDLC), The Role and Tasks of Systems Analyst, Attributes of a Systems Analyst, Tools Used by Systems Analyst, Approaches to Systems Development.

Unit III: Information Gathering and System Requirement specification

Hours: 11

Strategy to Gather Information, Information Sources, Methods of Searching for Information,

Interviewing Technique, Questionnaires, Other Methods of Information Search, System Requirements Specification, Data Requirements, steps in Systems Analysis, Modularizing Requirements Specifications, deciding on Project Goals, Examining Alternative Solutions, Evaluating Proposed Solution, Cost-benefit Analysis, Payback Period, Feasibility Report, System Proposal.

Unit IV: Data flow diagrams and Process Specification

Hours: 11

Symbols Used in DFDs, describing a System with a DFD, Levelling of DFDs, Levelling Rules,

Logical and Physical DFDs, Case Tools to Draw DFD, Process Specification Methods, Structured English, Decision Table Terminology and Development, Extended Entry Decision Tables, Eliminating Redundant Specifications

Unit V: Overview to Data input and designing output methods

Hours: 10

Data Input, Coding techniques, validating input data, interactive data input, objective of output designing, design of screens and output reports.

Unit VI: Implementation and maintenance of information systems

Hours: 09

Control in Information Systems, The Objectives of Control, Control Techniques, Audit of Information Systems, Security of Information Systems, designing reliable and

maintainable systems, design of software, software design and documentation tools, managing quality assurance, user training, training methods and conversion.

Core Books:

1. James A Senn: Analysis and Design of Information System, McGraw Hill International, 2003.
2. V. Rajaraman : Analysis and Design of Information Systems, Prentice-Hall of India Private Limited, 2003.
3. Kendall and Kendall: Systems analysis and Design, 5th Edition, Prentice-Hall of India Private Limited, 2003.

Reference Books:

1. Jeffrey L. Whitten, Lonnie D. Bentely and Kevin C. Dittman : Systems Analysis and Design Methods, Tata McGraw Hill Publishing Co. Ltd., 2001.
2. Tuthill and Leavy : Knowledge Based Systems : Mangers Perspectives : Tab professional and Reference Books, 1991.

Web references:

1. <http://lecture-notes-forstudents.blogspot.in/2010/04/system-analysis-anddesignsad.html>
2. <http://www.slideshare.net/aroravinay/1-introduction-to-ado-infosys>
3. <http://bcastuff.blogspot.in/p/sad-notes.html#.U6uW-ZzWleo>
4. <http://www.eis.mdx.ac.uk/staffpages/geetha/bis2030/DFD.html>
5. <http://faculty.washington.edu/ytan/is460/notes/LN11.pdf>

Course Outcomes: Upon successful completion of the course, students will:

C01	Understand basic concepts of system, types of information, qualities of information and its flow with respect to management structure
C02	Gain knowledge about how the system will develop and guide by well-experienced person in the organization with its documentation.
C03	Learn about various tools used to design flows of data, processes and data stores, performed by users of the system in graphical way.
C04	Understand various designing methods used to solve a problem while taking input and producing output as per given specification.
C05	Gain knowledge about various concepts of system maintenance and conversions

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basics of Information systems	✓				
2	Information system analysis Overview		✓			
3	Information gathering and System Requirement Specification		✓			
4	Data Flow Diagram and Process specifications			✓		
5	Overview to data input and design output methods				✓	
6	Implementation and maintenance of information systems					✓

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PSO1	PSO2
C01	3	3	2	2	2	3	3	2	1	3	2	3	2
C02	2	2	3	2	3	3	2	2	3	2	3	3	2
C03	2	2	3	3	1	2	2	1	2	1	1	3	1
C04	2	2	3	1	2	2	1	1	1	1	1	3	2
C05	2	2	2	2	1	2	2	1	1	1	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUC203: Programming the Internet

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components : Lab Assignments, Practical Test					

Prerequisites: Basic concepts of web.

Methodology, Pedagogy & Andragogy: This course focuses on providing hands-on experience to students for design and develop dynamic and interactive web sites using HTML5, CSS3 and JavaScript.

C. Outline of the Course:

Week No	Practical	Description
1	Basic Construction of an HTML & HTML5 Page, HTML Text Editors, HTML Building blocks. Adding Content: Basic HTML tags – Text formatting tags, heading tags, paragraph tags, division tag, Span tags, Break line tag, Binding space in HTML.	Student can learn basic concepts of web page design. Students get familiar with different tags and attributes which is used to add the text in a web page.
2	Controlling Font size and color, Adding Table – Table tag, grouping and merging table rows and columns. Adding List – ordered, unordered and definition list.	Student can able to change the font styles and color in more attractive way. Also, they can able to display the information in tabular & List format.
3	Adding Image, Adding links - Changing link colors, Email link, Embedding audio and video, Adding Field sets.	Student can add multimedia like image, audio, video in a web page. Student can explore various internal and external linking methods.
4	Adding Form: Form tag, Action, Method. Basic form controls – Textbox, Password field, Radio Button, Checkbox, Multiple line text area, file upload, Various types of buttons – submit, reset, image, simple button.	Student can design the various form elements for interaction with the web page users.
5	HTML5 input elements - Color, Number, Date, Month, week, Time, datetime-local,	Student can apply HTML5 input element to make form more attractive.

	Email, url, range, search,<datalist>, <output>, <progress>, <meter>	
6	Regular expression, HTML5 new attributes – Placeholder, Required, Pattern, Autocomplete, Autofocus, novalidate, formnovalidate, formaction, formmethod, spellcheck, contenteditable.	Student can explore the basic HTML5 new attributes and also apply validation on form submission.
7	CSS Styling the web - Applying CSS Building blocks –CSS Selectors, Cascade & inheritance, CSS Properties, CSS Box model.	Student can apply various types of CSS with its basic properties.
8	Creating Layout - Header, footer, navigation menu, and sub-navigation. Floats, Flexbox, Grids, Positioning, Multiple column layout.	Student can create different layout using semantic elements of HTML5 & CSS classification properties.
9	Creating Image Gallery and Animation - 2D/3D Transformation, Transitions and Animation, CSS Media query	Student can create layout image gallery & animation.
10	JavaScript – Adding Internal & External JavaScript, Working with variables, datatype – Number, String, Boolean.	Student can write script for basic programs using condition, Loop.
11	datatype –Array, Object, Operators, Conditional statements, Looping statements	Student can write script for basic programs using array and object.
12	Adding Dialog boxes in HTML page – Alert, Confirm and Prompt. Creating user defined function.	Student can able to use in-build and user defined functions of JavaScript
13	Adding Mouse and Keyboard Event in HTML page, Manipulating String and Date.	Students can make user interaction using various Events
14	JavaScript Document Object Model hierarchy – Create find and manipulate HTML Element using Objects and methods	Student can explore the DOM Hierarchy.
15	Form validation, Applying Style using JavaScript. Creating New window, Accessing & manipulating History of HTML Pages.	Student can apply various objects to make web page dynamic with user interactions.

Total hours: 60

❖ **Text Books:**

1. Matthew MacDonald: HTML5 The Missing Manual, O'Reilly Media, August 2011.
2. Peter Gasston: The Book of CSS3, A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.

3. Richard York: Beginning CSS Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

Reference Books:

1. Ivan Bayros : Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, fourth revised edition , BPB Publication.
2. David Mc Farland : CSS The Missing Manual, O'Reilly, 2006.
3. Julie C. Meloni : HTML, CSS and JavaScript All in One, Pearson.

Web References:

1. <http://www.tutorialspoint.com/html5> [For notes on HTML5 tags]
2. <http://www.w3schools.com/html/> [For HTML5, CSS and JavaScript notes and examples]
3. <https://in.godaddy.com/help/dreamweaver-cs6-publish-your-website-7811>[Publish your website using Dreamweaver]
4. <http://fullbooksfreedownload.blogspot.in/2016/02/html-css-javascript-web-publishing-in.html> [Book :- HTML, CSS & JavaScript Web Publishing in One Hour a Day, Sams Teach Yourself, 7th Edition PDF]

Course Outcome: Upon successful completion of the course, student will:

C01:	Able to understand basic construction of HTML & HTML5 document and started using basic tags.
C02:	Able to understand the implementation of HTML tags and able to create HTML static Web pages.
C03:	Able to use basic properties of CSS and CSS3 in html file and Able to create HTML, CSS layout.
C04:	Able to write client side script – JavaScript to perform basic program.
C05:	Able to use DOM objects for dynamic web pages.

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	2	1	-	2	-	-	-	2	-	1	1
C02	3	2	2	1	-	2	-	-	-	2	-	1	1
C03	3	3	3	1	-	2	-	-	-	2	-	2	1
C04	3	3	2	1	-	2	-	-	-	2	-	2	1
C05	3	3	2	1	-	2	-	-	-	2	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUS201: Computer Oriented Numerical Methods

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The course covers key topics like finding roots of equations, solving linear equations, interpolation, and solving differential equations. Teaching includes lectures, interactive problem-solving, and project-based assessments with continuous feedback. The focus is on practical applications and real-world problems, using flexible and engaging teaching methods. The aim is to make the content relevant and useful for students' careers and interests.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Numerical Methods	08
2	Interpolation	09
3	Numerical Integration and Differentiation	10
4	Solution of Linear Equation	12
5	Solution of Differential Equations	11
6	Solution of Partial Differential Equations	10

Total Hours (Theory): 60
Total Contact Hours: 60

Detailed Syllabus:

Unit - I: Numerical Methods

Hours: 08

Introduction to Numerical Methods, Various methods such as Bisection, False-Position and Newton Raphson methods, Graffes Root Squaring Method, Convergence of Solution

Unit - II: Interpolation

Hours:09

What is an interpolation, Difference tables, Newton forward and backward Interpolation formula, Lagrange's formula, Newton's Divided Difference Formula

Unit - III: Numerical Integration and Differentiation

Hours:10

Numerical Integration, Trapezoidal rule, Differentiating Continuous function (Two point and Three-point formula), Tabulated function and its maxima and minima, Newton-Cote's Formula.

Unit - IV: Solution of Linear Equation

Hours:12

Solution of Linear Equation, Gauss-Jordan Elimination methods, Gauss - Seidal and Gauss Jacobi Interactive methods

Unit - V: D Solution of Ordinary Differential Equations

Hours:11

Solution of Ordinary Differential Equations, Taylor Series and Euler Methods, Runge-Kutta Methods.

Unit - VI: Solution of Partial Differential Equations:

Hours:10

Review and examples of partial differential equations, classification of partial differential equations, Difference equation, Laplace's equation and Poisson's equation.

Core Books:

1. D. N. Datta: Computer Oriented Numerical Methods, Vikas Pub. House, 2012.
2. S.S. Sastry : Introductory methods of Numerical Analysis, Prentice Hall of India, 4th edition 2006

Reference Books:

1. V. Rajaraman Computer Oriented Numerical Methods, 3rd Edition, Prentice-Hall of India Pvt.Ltd, 3 rd edition, 2016
2. M. Ray, Har Swarup Sharma, Sanjay Chaudhary: Mathematical Statistics, Ram Prasad and sons, 11th edition
3. B S Grewal: Numerical Methods In Engineering & Science (With Programs In C, C++ And MATLAB), Khanna Publishers, 10th edition, 2014

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	Able to use Numerical methods in computer technology
C02 :	To learn Interpolation
C03 :	To Learn the concepts of Numerical Integration and Numerical Differentiation
C04 :	To understand linear equations
C05 :	To learn techniques to solve ordinary differential and partial differential equations

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Numerical Methods	√				
2	Interpolation		√			
3	Numerical Integration and Differentiation			√		
4	Solution of Linear Equation				√	
5	Solution of Differential Equations					√
6	Solution of Partial Differential Equations					√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	3	2	1	-	2	-	-	-	2	-	2	1
C02	3	3	2	1	-	2	-	-	-	2	-	2	1
C03	3	3	2	1	-	2	-	-	-	2	-	2	1
C04	3	3	2	1	-	2	-	-	-	2	-	2	1
C05	3	3	2	1	-	2	-	-	-	2	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUS202: Computer Oriented Management System

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/C ontact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The theory sessions will emphasize real-time systems. To enhance comprehension, we will provide illustrations using actual data across various systems. The sessions will meticulously explain the system's functionality through detailed flow descriptions. To bridge the gap between theoretical knowledge and practical application, we have to schedule industrial visits. These visits are designed to offer students a tangible correlation between their studies and real-world practices. Students should be assigned case studies to facilitate the practical application of their acquired knowledge. These case studies aim to deepen understanding and foster the ability to apply theoretical concepts to real-world scenario.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Basics of Information System	09
2	Architecture and Design of Information System	08
3	Transaction Processing System	09
4	Management Information System	06
5	Decision Support System & Executive Support System	18
6	Introduction to ERP and trending technologies	10

Total Hours (Theory): 60

Total Contact Hours:60

Detail Syllabus:

Unit I : Basics of Information System

Hours 09

Role of data and information, Types of Systems, Types of information, Quality of Information, what is an Information System, Functions of Information System, classification of an Information System, Major Types of Information Systems, Interrelationship of Information Systems.

Unit II : Architecture and Design of Information System

Hours 08

Meaning of Architecture, development and maintenance of information system, centralized and decentralized systems, factors of success and failure in information system and risk involved with information system.

Unit III : Transaction Processing System

Hours 09

Introduction of Transaction Processing System (TPS), Transaction Processing System (TPS) Architecture, Transaction Processing System (TPS) Life cycle, Methods of Transaction Processing System (TPS), Information System Availability Control, Illustrations and live demonstration of TPS such as Banking System, Railway reservation system etc.

Unit IV : Management Information System

Hours 06

Introduction to Management: Approaches of Management, Function of Management, System from Functional Perspective, Reports of Management Information System, A business perspective of MIS, Dimensions of MIS, contemporary Approaches to Information System.

Unit V : Decision Support System and Executive Support System

Hours 18

Business value of Improved Decision making, Types of Decision, Decision making Process, The difference between MIS and DSS, Components of DSS, System for Decision Support Group Decision Support System, Business value of GDSS. Important Dimensions of knowledge, Organizational learning and Knowledge Management, The Knowledge Management value chain, Overview of different types of Knowledge Management Systems, Characteristics of ESS, The Role of ESS in the Firm, Business value of ESS, Monitoring Corporate Performance, Working Examples of ESS.

Unit VI : Introduction to ERP and trending technologies

Hours 10

An overview of ERP, Basic ERP Concepts, Risk and Benefits of ERP, ERP and related technologies, Business Intelligence, Some case studies on ERP or Information system.

Core Books:

1. K. C. Laudon and J. P. Laudon: Management Information Systems, 12th Edition, Pearson Education.
2. Alexis Leon: ERP Demystified, Second Edition, Tata McGraw-Hill.

Reference Books:

1. W.S. Jawadekar: Management Information Systems, 2nd Edition, Tata McGraw-Hill.

Web References:

1. <http://www.slideshare.net/NorazilaMat1/laudon-mis12-ppt01-16595885>
[Function of Information system, A business perspective of MIS, Dimensions of MIS, Contemporary Approaches to Information System]
2. <http://www.uh.edu/~mrana/try.htm> [Types and Functions of Information system]
3. <http://bisom.uncc.edu/courses/info2130/Topics/istypes.htm> [Types of Information system]
4. <http://kalyan-city.blogspot.com/2011/05/levels-of-management-top-middle-and.html> [classification of information system]
5. [http://my.safaribooksonline.com/book/management/9780470916803/operational-planning-and-control-systems/management levels comma functions comm](http://my.safaribooksonline.com/book/management/9780470916803/operational-planning-and-control-systems/management%20levels%20comma%20functions%20comm)
[classification of information system]
6. <http://ambarwati.dosen.narotama.ac.id/files/2011/05/FIS-2011-w2.pdf>
[Transaction processing system]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand basic of data & information, types of information system, function of information system and its quality.
CO2 :	To understand designing and architecture of information system
CO3 :	To understand transaction processing concepts and its working.
CO4 :	Learning Management Information system, its types and its implementation. To gain understanding of Decision Support Systems.
CO5 :	To learn Knowledge Management, Executive Support Systems, ERP and its technology.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basics of Information System	√				
2	Architecture and Design of Information System		√			
3	Transaction Processing System			√		
4	Management Information System				√	
5	Decision Support System & Executive Support System				√	
6	Introduction to ERP and trending technologies					√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
CO1	1	3	1	-	3	-	1	2	3	3	-	1	1
CO2	1	3	1	-	2	1	-	2	3	3	2	-	1
CO3	1	3	1	1	3	3	-	2	3	3	-	-	1
CO4	1	2	1	-	2	3	1	2	2	2	1	1	1
CO5	1	3	1	-	3	3	-	1	3	3	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUA201: Life Management

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	02	30	25	55
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of life management. Students will be introduced life styles and its basic, time management, work efficiency, yoga, physical health management, stress and mental health management, and personality skills. Students will give experimental learning exposure in form of assignment and case study.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Life Style and its Basics	06
2	Work Efficiency	04
3	Personality Skills	06
4	Time Management for students	05
5	Physical health management	04
6	Stress and mental health management	05

Total Hours (Theory):30
Total Contact Hours:30

Detail Syllabus:

Unit I : Life Style and its Basics

Hours 06

Purpose of life and its dimensions, Importance of Self-Evaluation; (Daily routine, Food habits, Dressing Sense, Habit formation, Company, Etiquettes), Duties and Commitment of Self, Family and Society, Adjustment with Self and Environment

Unit II : Work Efficiency

Hours 04

Positive way of thinking, Tools & techniques for Positive thinking, Karma and Karma Phal Sidhanta, Behavioural Skill

Unit III : Personality Skills

Hours 06

Self-Assessment Techniques, Adjustment Skills, Art of Positive Thinking, Developing Self Confidence,

Unit IV : Time Management for students

Hours 05

Importance, Steps, Barriers and solutions of Time Management, Reading and Writing Skill, Making Notes and Conceptual Clarity of Subject topic.

Unit V : Physical health management

Hours 04

Importance of physical health and exercise, importance of yoga, physical problems, exercise for health improvements

Unit VI : Stress and mental health management

Hours 05

Importance of mental health, spirituality, improving concentration, importance of Dhyan, Memory problem and Memory Boosting techniques, causes and effects of stress, importance of Prathana, Upasna, Sadhna, and Aradhna

Core Books:

1. Time Management, McGraw-Hill, 2003
2. How To Manage Your Life, Vishal Tatwaved, 2012.

Reference Books:

1. Living Stress Free - The Secret of How to Manage Stress and Live Life Fully, Sonali Perera, SP MEDIA MARKETING.
2. TIME MANAGEMENT, BRIAN TRACY, American Management Association, 2013.

Web References:

1. Life management: a holistic approach to make the most of your life, Dr. Hannah Rose <https://nesslabs.com/life-management>
2. Life Management, <https://www.sciencedirect.com/topics/computer-science/life-management>
3. <https://edynamiclearning.com/course/health-1-life-management-skills/>

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	Students will learn basics of life managements
C02 :	The students will be familiar with various styles
C03 :	Students will get the knowledge of time management and stress management
C04 :	Students will get various aspects of physical and mental health
C05 :	They will be able to develop personality and good habits

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Life Style and its Basics	√				
2	Work Efficiency		√			
3	Personality Skills					√
4	Time Management for students			√		
5	Physical health management				√	
6	Stress and mental health management				√	

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	1	3	1	-	3	-	1	2	3	3	-	1	1
C02	1	3	1	-	3	1	-	1	3	3	1	-	1
C03	1	3	1	1	3	3	-	2	3	3	-	-	1
C04	1	2	1	-	2	3	1	2	2	2	1	1	1
C05	1	3	1	-	3	3	-	1	3	3	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**TEACHING SCHEME &
DETAILED SYLLABUS**

FOR

**BSc-IT PROGRAMME
(4th SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2024-25**

B.Sc. (IT) Semester-IV

Course Code	Course Name	Theory / Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CTUC204	Fundamentals of Operating Systems	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUC205	Fundamentals of Visual Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CTUE201 CTUE202	(Select any one) 3. Introduction to Artificial Intelligence 4. Computer Network Security	Theory	50	50	-	-	04	Minor
CTUA202	Social Media and Blog Writing	Practical	-	-	25	25	02	AEC
	University Elective-II	Practical	-	-	50	50	02	MDC
HSUV203	Indian Knowledge System	Practical	-	-	25	25	02	VAC
CAUT201	Summer Internship and Viva - II	VIVA	-	-	50	50	04	Exit Course
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.)								
*SEE- Semester –End- Evaluation								

University Elective - II			
Sr. No.	Course Code	Course Name	Department/Faculty
1		Prototyping Electronics with Arduino	Engineering
2		Web Designing	Engineering
3		Basics of Environmental Impact Assessment	Engineering
4		Internet Technology & Web Design	Engineering
5		Material Science	Engineering
6		Cosmetics in Daily Life	Pharmacy
7		Life Style Diseases & Management	Nursing
8		Occupational Health & Ergonomics	Physiotherapy
9		Health Care Management	Management
10		Astrophysics, Space and Cosmos-2	Applied Science
11		MATLAB Programming	Engineering
12		Maintenance of Household Apparatus	Engineering
13	CAUD202	Programming the Internet	Computer Science

CTUC204 : Fundamentals of Operating Systems

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory) : Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical) : Lab Assignments, Practical Test, VIVA					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: In order to achieve the course objectives, students will be introduced to the basic operating system concepts and basic functions. Students will be offered appropriate case studies in order to provide actual exposure to theoretical operating system ideas. During the practical session, student will commence the learning from MS-DOS commands in windows environment. The student will also get hands on working with basic Linux commands. Finally using base of Linux commands, shell scripts would be written to demonstrate operating system programming.

Outline of the course (Theory):

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Operating System Fundamentals	09
2	Memory Management	11
3	Processor Management	11
4	Device Management	11
5	File Management	10
6	Introduction to Shell Script	09

Total Hours (Theory): 60
Total hours (Practical): 60
Total Contact Hours:120

Detail Syllabus:

Unit I: Operating System Fundamentals

Hours 09

What is an operating system, Brief history of an operating system, Pyramid structure of an operating system, Operating system hardware and software, Evolution of an operating system, types of operating system, Batch system, Interactive system, Real time system, Hybrid system, Shell, compiler, assembler, interpreter.

Unit II: Memory Management

Hours 11

Memory management of early system, Fixed partition, Dynamic partition, Re-locatable dynamic partition, Best fit and first fit allocation method, De-allocation, Paged memory allocation, Demand paging allocation, segmented memory allocation, Virtual memory, cache memory, Associative memory, Page replacement policies.

Unit III: Processor Management

Hours 11

What is job and Process, Multiprogramming concept, Process scheduler, Job scheduler, various process states, Thread control, Process Control Block (PCB), Processor scheduling policies, FCFS, SJN, SRT, RR, EDF, Multiple level queues.

Unit IV: Device Management

Hours 11

Meaning, Depreciation of money, Monetary inflation concepts, types of inflation, Causes and effect of inflation different sectors of the economy, Causes of Inflation, Advantages, Formula and Disadvantages of inflation.

Unit V: File Management

Hours 10

Fundamentals of File management, File naming conventions, Fixed length and variable length records, Physical file organization, Physical storage allocation, File access methods, File control matrix, File compression.

Unit VI: Introduction to Shell Script

Hours 09

Shell Script, Shell environment, Making Script interactive, Command Line Argument, Evaluating Expression, Operator, Control Statement, Looping statement.

Outline of the Course (Practical):

Week No	Practical	Description
1	Disk operating system -Commands	Introduction to MS DOS environment with basic commands
2	Introduction to Linux Editor	Introduction to Cygwin as Linux Editor along with its usage
3	Linux Operating System-Commands	Using basic commands of Linux
4	Linux Operating System-Advanced Commands	Implementation of advanced Linux commands used in industry
5	Vi editor-Create Text file	Practice of writing and modifying various files in Vi editor
6	Introduction of Shell Programming-Create .sh file and Execute file	Creating simple shell script to get introduced with operating system programming
7	Variable and Rules for variable	Creating and using variables, and constants in shell scripts
8	Interactive Shell Script, Working with Command Line Arguments	Practicing shell scripts which are responsive to the user input
9	Working with expression	Arithmetic expressions in shell script
10	Conditional statement-IF, Working with Operator	Learning conditional statements such as if, elif, case. Using arithmetic, relational, and logical operators.
11	Looping Statement-While	Writing shell scripts with while loops with real life examples
12	Looping Statement-Until, For	Writing shell scripts with until and for loops with real life examples
Total hours: 60		

Core Books:

1. Ann McIver McHoes, Ida M Flynn: Understanding Operating System, 7 Edition, Cengage Learning, 2014.
2. William Stallings: Operating Systems Internals and Design Principles, 5th Edition, PHI, 2005.
3. Sumitabha Das: Unix concepts & application, 4th Edition, Tata McGraw Hill, 2010.
4. Kenneth Rosen, Douglas Host, James Farber and Richard Rosinski: The Complete Reference, Tata McGraw Hill, 1999.
5. Christopher Negus: Linux Bible, 10th Edition, Wiley, 2020.

Reference Books:

1. Silberschatz: Operating System Concepts, 5th Edition, John Wiley & Sons (ASIA) Ltd., 2008.
2. Mark G. Sobell: A Practical Guide to Linux, Pearson Education, 1997.
3. KJ.George, Operating System Concepts and Principles, Sroff Publishers, 2003.

Web References:

1. <https://study.com/learn/operating-system.html> [Operating System Video Lessons]
2. <https://www.studytonight.com/operating-system/> [Operating System Tutorial]
3. <http://www.ics.uci.edu/~ics43/lectures.html> [Lecture notes of OS]
4. <http://www.tutorialspoint.com/unix/index.htm> [Unix Tutorials]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand the fundamental concepts of an operating System
CO2 :	To understand memory allocation and de-allocation methods
CO3 :	To study fundamentals of processor and its scheduling policies
CO4 :	To study various supported device and file access technologies
CO5 :	To understand the commands of disk operating system, Linux operating system and Learning basic shell programming.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Operating System Fundamentals	√				
2	Memory Management		√			
3	Processor Management			√		
4	Device Management				√	
5	File Management				√	
6	Introduction to Shell Script					√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	1	1	2	2	-	2	2	-	1	1	-	3	2
C02	3	2	2	2	-	2	2	-	1	1	-	3	2
C03	3	2	2	2	-	2	2	-	1	1	-	3	2
C04	3	2	2	2	-	2	2	-	1	1	-	3	2
C05	2	2	3	3	-	2	3	-	1	1	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUC205 : Fundamentals of Visual Programming

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory) : Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical) : Lab Assignments, Practical Test, VIVA					

Pre-requisite: Basic knowledge of C & C++ programming language.

Methodology, Pedagogy & Andragogy: During theory lectures illustrations emphasizing the need for advanced features of .Net framework and to develop windows and web-based application using C# language will be given. During practical sessions, students will actively engage in developing both desktop and web applications, applying the concepts discussed in lectures as well as will explore various features of Visual Studio and .NET.

Outline of the course (Theory):

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1.	Introduction to .NET framework	09
2.	Basics of Object Oriented programming using C#	11
3.	Building desktop application	11
4.	Web programming fundamentals	10
5.	Database connectivity through ADO.NET	11
6.	Building WPF application	08

Total Hours (Theory): 60

Total hours (Practical): 60

Total Contact Hours:120

Detail Syllabus:

Unit I: Introduction to .NET framework

Hours 09

Introducing .NET Framework, .NET Framework Component, .NET Framework Version Compatibility, Core of .NET Framework: Application Services, Base Class Library and CLR, Types of Applications which can be developed using MS.NET, MS.NET Base Class Library, MS.NET Namespaces, The Common Language Runtime (CLR), Managed Code, MS.NET Memory Management / Garbage Collection, Common Type System (CTS), Common Language Specification (CLS), Types of JIT Compilers

Unit II: Basics of Object-Oriented programming using C#

Hours 11

Overview of Object-Oriented Programming, Principles of Object Orientation programming: Encapsulation, inheritance, polymorphism, Object, Defining Properties, Using Auto-Implemented Properties, Defining Methods, Understanding Constructors, Extending .NET Classes via Inheritance, Defining and Implementing Interfaces, Understanding the Role of Interfaces in .NET. C# Datatypes & Variables Declaration, Implicit and Explicit Casting, Checked and Unchecked Blocks –Overflow Checks, Casting between other datatypes, Boxing and Unboxing, Enum and Constant, Operators, Control Statements, Working with Arrays, C# Application Basics: Command line and VS.NET compilation.

Unit III: Building desktop application

Hours 11

Building Windows Forms Applications, Setting Form Properties, Understanding the Life-cycle of a Form, Using the Windows Forms Designer, MessageBox Class and .config File, Working with Windows Forms Controls: TextBox, Button, Selection, List, Container, Image, ErrorProvider, TooltipProvider, etc. Handling Events: Understanding the Event-Driven Programming Model, Writing Event Handlers, Sharing Event Handlers.

Unit IV: Web programming fundamentals

Hours 10

Basic of Http Request and Http Response. o Understand form Tag and Comparison between Get and Post. ASP.NET Page Life Cycle. Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config, Rich Web Control, Validation Controls, Concept of custom control.

Unit V: Database connectivity through ADO.NET

Hours 11

Evolution of ADO.NET, Understanding the ADO.NET Object Model, Connected vs. Disconnected Access, Use of Connection, Command, DataReader, DataAdapter and Dataset class.

Unit VI: Building WPF application

Hours 08

Overview of WPF, Features of WPF, Windows Forms vs WPF, Introduction to XAML, WPF Layout: Grid Panel, Stack Panel, Dock Panel, Wrap Panel and Canvas Panel, WPF Controls: TextBlock, PasswordBox, ListBox, Slider, Popup, Menus and Expander

Outline of the Course (Practical):

Week No	Practical	Description
1	Basics of Visual Studio IDE and .NET Introduction	Getting familiar with Visual Studio IDE and basic programming structure of C#.
2	Simple C# Programs	Writing basic C# programs, understanding syntax, and compiling with Visual Studio.
3	Console Applications	Developing console-based applications, using input and output statements, and understanding the main method.
4	Object-Oriented Programming Concepts	Implementing encapsulation, inheritance, and polymorphism in C#.
5	Control Statements and Arrays	Using operators, control statements, and working with arrays in C#.
6	Building Windows Forms Applications	Working with TextBox, Button, and other common controls in Windows Forms.
7		
8		
9	Event Handling in Windows Forms	Understanding event-driven programming and writing event handlers.
10	Basics of ASP.NET	Understanding HTTP requests and responses, and creating a simple ASP.NET page.
11		
12	Database Connectivity with ADO.NET	Connecting to a database, executing commands, and retrieving data using ADO.NET.
13		
14	A Fully Windows Form Application	Designing and implementing a complete Windows Forms application, integrating various controls, handling events, and database connectivity.
15	Building a WPF Application	Creating a basic WPF application.

Total hours: 60

Core Books:

1. Christian Nagel : Professionals C# and .NET 2021 Edition , Wiley – India, WROX
2. Albahani Josheph: C# 4.0 Pocket Reference, Shroff publication,2010.

Reference Books:

1. Jon Skeet, C# in Depth, Managing Publication,2010
2. Drayton David: C# Language Pocket Reference, Shroff publication,2005

Web References:

1. <http://csharp.net-informations.com/> [to learn C# and .NET fundamentals]
2. <http://www.wpftutorial.net/>[to learn WPF]
3. <https://dotnet.microsoft.com/en-us/learn/csharp> [tutorial and videos to learn C#]
4. <https://dotnettutorials.net/course/csharp-dot-net-tutorials/>
5. <https://www.javatpoint.com/c-sharp-tutorial>

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	Understanding .NET Framework core features and Components.
C02 :	To develop object oriented programming skills using C#.
C03 :	To develop windows and web based application using core feature of .NET framework and C# language
C04 :	Access and manipulate data in a Microsoft SQL Server database by using MicrosoftADO.NET.
C05 :	Use modern approach to build windows based application using WPF

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to .NET framework	√				
2	Basics of Object Oriented programming using C#		√			
3	Building desktop application			√		
4	Web programming fundamentals			√		
5	Database connectivity through ADO.NET				√	
6	Building WPF application					√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02
C01	3	2	2	3	-	2	1	-	-	1	1	2	-
C02	2	3	2	2	-	3	2	-	-	2	-	1	-
C03	2	3	3	2	-	2	2	-	-	2	1	2	1
C04	2	3	3	2	-	3	3	-	-	2	2	2	1
C05	-	3	3	-	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUE201 : Artificial Intelligence (AI) Fundamentals

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/C ontact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The theory sessions will be focused on basics of Artificial Intelligence, problem solving paradigms, and search strategies. Areas of application such as Expert Systems, Software Agents, knowledge representation, natural language processing, expert systems, robotics will be explored.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Artificial Intelligence	08
2	Problem-solving by Search and constraint satisfaction	11
3	Knowledge representation and reasoning	11
4	Expert Systems	11
5	Agents	10
6	Introduction to AI Techniques and Application Areas	09

Total Hours (Theory): 60

Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to Artificial Intelligence

Hours 08

Concepts and definitions of Artificial Intelligence (AI) – Brief history of AI – AI and related fields – The AI problems and underlying assumptions, AI approaches: Cognitive Science, Turing Test.

Unit II: Problem solving by Search and constraint satisfaction

Hours 11

Problem spaces; brute-force search; best-first search; two-player games; constraint satisfaction

Unit III: Knowledge representation and reasoning

Hours 11

Review of propositional and predicate logic; resolution and theorem proving; non-monotonic inference; probabilistic reasoning; Bayes theorem, reasoning under uncertainty.

Unit IV: Expert Systems

Hours 11

Introduction – Representing and using domain knowledge – Knowledge acquisition and representation – General structure of Expert Systems – Expert System Shell – Advantages and disadvantages of Expert Systems, Applications of Expert Systems

Unit V: Agents

Hours 10

Definition of agents; successful applications and state-of-the-art agent-based systems; software agents, personal assistants, and information access; multi-agent systems.

Unit VI: Introduction to AI Techniques and Application Areas

Hours 09

Introduction to fuzzy logic – Introduction to various application areas of AI like Natural Language Processing (NLP), Learning, Speech Recognition, Computer Vision, Game Playing, Robotics.

Core Books:

1. Norvig, Peter, Paradigms of AI Programming, Morgan Kauffman, 1992.
2. S. Russell and P. Norvig, Modern Approach to Artificial Intelligence, Prentice Hall of India Ltd., 2006.
3. Rich & Knight, Artificial Intelligence, 2nd edition, TMH, 1991.

Reference Books:

1. R.Akerker and P. S. Sajja, Knowledge-Based Systems, Jones and Bartlett, MIT, 2010.
2. George Luger, Artificial Intelligence, 5th Edition, Addison Wesley, 2004.

Web References:

1. http://www.tutorialspoint.com/artificial_intelligence/ [Basics of AI]
2. <http://intelligence.worldofcomputing.net/ai-branches/expert-systems.html> [Expert Systems]
3. <http://pages.cs.wisc.edu/~bolo/shipyard/neural/local.html> [Neural Network]

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	To understand basic problems of AI and history of AI.
C02 :	Getting deep knowledge of searching algorithms and its applications in AI.
C03 :	Learning Knowledge Representation and reasoning techniques.
C04 :	To gain understanding Expert System and Agent Based Systems.
C05 :	To explore various application areas of AI.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Artificial Intelligence	√				
2	Problem solving Search and constraint satisfaction		√			
3	Knowledge representation and reasoning			√		
4	Expert Systems				√	
5	Agents				√	
6	Introduction to AI Techniques and Application Areas					√

Course Articulation Matrix:

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P10	P11	PS01	PS02
C01	-	2	2	2	1	-	-	-	-	-	-	1	2
C02	3	2	2	3	2	2	-	3	-	-	-	2	2
C03	-	-	3	3	2	-	-	-	-	-	-	1	-
C04	3	2	2	-	2	2	-	3	-	-	-	1	2
C05	2	-	-	-	2	-	3	3	-	-	-	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUE202: Computer Network Security

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study, Certification courses, etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The methodology for teaching Computer Network and Security involves a hybrid approach that blends theoretical knowledge with practical application, catering to both novice and experienced learners through pedagogy and andragogy. This includes the use of multimedia resources, simulations, and hands-on exercises to facilitate understanding. Andragogically, the course engages adult learners by emphasizing problem-based learning, real-world case studies, and collaborative projects that draw on learners' prior knowledge and professional experiences. This approach ensures learners can critically analyze and apply security measures in complex networking environments, adapt to emerging threats, and design secure network infrastructures.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Computer Network Security	07
2	Symmetric Ciphers	11
3	Public Key Cryptography	12
4	Authentication and Authorization	11
5	Network Security	10
6	Web Security	09

Total Hours(Theory):60

Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to Computer Network Security

Hours 07

Need for Security, Security Attacks, Services and Mechanisms, Network Security, Models

Unit II: Symmetric Ciphers

Hours 11

Substitution & Transposition Techniques, Block Cipher, DES, Triple DES, Stream Ciphers, RC4

Unit III: Public Key Cryptography

Hours 12

Need and Principles of Public Key Cryptosystems, RSA Algorithm, Key Distribution and Management

Diffie-Hellman Key Exchange, Digital Signatures

Unit IV: Authentication and Authorization

Hours 11

Authentication Requirements, Message Authentication Codes, Hashes, MD5 & SHA, User Authentication: Password, Certificate based & Biometric Authentication 6. Kerberos

Unit V: Network Security

Hours 10

Firewalls, IP Security, VPN, Intrusion Detection, SSL, TLS

Unit VI: Web Security

Hours 09

Web Security, Security Principles (CIA Triad: Confidentiality, Integrity, Availability), Vulnerability Scanning (e.g., OWASP ZAP, Burp Suite), Content Security Policy (CSP)

Core Books:

1. Kizza JM. Computer network security. Springer Science & Business Media; 2005 Apr 7.
2. Wang J. Computer network security. Berlin/Heidelberg, Germany: Springer; 2009.

Reference Books:

1. Buchmann J. Introduction to cryptography. New York: Springer; 2004 May.
2. Delfs, H., Knebl, H., & Knebl, H. (2002). *Introduction to cryptography* (Vol. 2). Heidelberg: Springer.

3. Garfinkel S, Spafford G. Web security, privacy & commerce. " O'Reilly Media, Inc."; 2002.
4. Garfinkel, Simson, and Gene Spafford. *Web security & commerce*. Cambridge, MA: O'reilly, 1997.

Web References:

1. <https://www.cisco.com/c/en in/products/security/what-is-network-security.html>
2. <https://www.javatpoint.com/computer-network-security>
3. <https://tndalu.ac.in/econtent/9 Computer Network And Network Security.pdf>

Course Outcomes: Upon successful completion of the course, students will be,

C01 :	Develop Concept of Security needed in Communication of data through computers and networks along with Various Possible Attacks
C02 :	Understand Various Encryption mechanisms for secure transmission of data and management of key required for required for encryption
C03 :	Understand authentication requirements and study various authentication mechanisms
C04 :	Understand network security concepts and study different Web security mechanisms.
C05 :	Understand web security concepts

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Computer Network Security	√				
2	Symmetric Ciphers		√			
3	Public Key Cryptography			√		
4	Authentication and Authorization				√	
5	Network Security					√
6	Web Security					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PS01	PS02
C01	2	1	1	-	2	1	-	-	1	-	2	2	2
C02	1	2	2	2	3	1	-	-	1	1	2	2	2
C03	1	2	2	2	2	2	1	-	1	-	2	2	2
C04	2	2	2	2	3	3	2	-	2	1	3	3	3
C05	1	2	2	1	2	2	1	-	1	1	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUA202: Social Media and BLOG Writing

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/C ontact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components : Lab Assignments, Practical Test, VIVA					

Prerequisites: None.

Methodology, Pedagogy & Andragogy: This course focuses on providing hands-on experience to students for understanding social media marketing strategies and BLOG writing skills.

Outline of the Course:

Week No	Practical	Description
1	Introduction to Social Media - Platforms for social media marketing - Roles of social media in digital marketing	Learn basics of Social Media
2	Optimizing for Social Media Platforms - Facebook - Instagram - X (formerly Twitter)	Learn about various social media platforms and optimizing for them.
3	Content Calendar for Social Media Creating content calendar with different focus	
4	Content creation and Management in Meta Business Suite Understanding and Managing Meta Business Suite	Understanding Search Engine Result Page, and learn about On-the-Page and Off-the-Page Optimization
5	Paid Marketing in Meta Platforms – I - Understanding Ads Manager - Understanding Billing	Understanding paid marketing in Meta platforms
6	Paid Marketing in Meta Platforms – II - Brand awareness campaign - Traffic campaign - Engagement campaign	Understanding paid marketing in Meta platforms with different objectives
7	Paid Marketing in Meta Platforms – III - Lead generation campaign - Conversion campaign	Understand to embed Social Media with website

8	Getting data from Social Media Paid Advertisement using Meta Pixel	For custom audiences and lookalike audiences
9	Introduction to BLOG • What is BLOG? • Importance of BLOG	-
10	Popular Blogging Platforms • WordPress • Blogger	-
11	Using SEO in Blogging • Using AI in Blogging • Understanding SEO in Blogging	-
12	Using Blogger for Blogging	-
13	Using WordPress for Blogging	-
14	Case Study – I (Social Media)	-
15	Case Study – II (Blogging)	-
Total hours: 60		

Text Books:

1. Damian Ryan and Calvin Jones: Understanding Digital Marketing, Kogan Page, 2014
2. Stephanie Diamond: Digital Marketing All-in-One for Dummies, Wiley Publication, 2019
3. Amy Lupold Bair, Blogging for Dummies, 7th Edition, Wiley Publication, 2020

Reference Books:

1. Philip Kotler, Hermawan Kartajaya, and Iwan Setiawan: Marketing 4.0: Moving from Traditional to Digital, Wiley Publication, 2016
2. Jan Zimmerman and Deborah Ng: Social Media Marketing All-in-One for Dummies, Wiley Publication, 2017

Web References:

1. <https://www.simplilearn.com/tutorials/digital-marketing-tutorial>
2. <https://www.tutorialspoint.com/digital-marketing/index.htm>
3. <https://intellipaat.com/blog/digital-marketing-tutorial/>
4. <https://www.javatpoint.com/digital-marketing>
5. <https://www.javatpoint.com/blog>

Course Outcome: Upon successful completion of the course, student will:

C01:	Able to understand digital marketing and its importance.
C02:	Able to create and manage content using social media platforms
C03:	Able to understand paid advertisements on social media platforms
C04:	Able to blogging and its importance in digital marketing.
C05:	Able to use generative AI in Digital Marketing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	1	2	1	3	2	2	1	1	-	1	1	1	1
C02	-	1	2	3	1	3	1	-	-	2	2	2	2
C03	1	1	2	3	1	3	1	-	-	2	2	2	2
C04	1	1	2	3	1	1	1	1	-	1	-	1	1
C05	2	2	3	3	2	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CTUI201 : Summer Internship and Viva – II

Objective and Scope:

Students who choose to exit the programme after the completion of the first two semesters must undertake internships, community engagement programmes and field based learning/minor projects during the summer term. This internship, worth 4 credits, is a requirement for the award of the Diploma. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. This provision ensures that students gain valuable practical experience and industry exposure, complementing their academic learning and enhancing their career prospects. Following are the intended objectives of internship training:

- **Practical Experience:** Provide hands-on experience with real-world projects to apply theoretical knowledge gained during the course.
- **Skill Development:** Enhance technical skills, such as programming, software development, database management, and other relevant IT skills.
- **Industry Exposure:** Familiarize students with the working environment of the IT industry, including understanding company culture, workflows, and professional practices.
- **Professionalism:** Instil a sense of professionalism and work ethic, including adhering to deadlines, following instructions, and working collaboratively.
- **Transition to Employment:** Facilitate a smoother transition from academic life to professional careers by bridging the gap between theory and practice.

Guidelines of Internship:

The Summer Internship shall be of 15 (Minimum 60 hours) duration and will be undertaken during the Summer Vacation. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. It is mandatory for the students to seek written approval from the coordinator about the topic and the organization before commencing the Summer Internship. During Summer Internship students are expected to take necessary guidance from the faculty guide allotted by the Institute. To do it effectively they should be in touch with their guide through e-mail or phone. Students must maintain regular reports detailing the work completed, challenges faced, and learning experiences. Students must submit a final report and presentation summarizing the internship experience, project outcomes, and key learnings to guide. The technical specifications for report preparation is as under:

Technical details:

1. The report shall be printed on A-4 size white paper.
2. 12 pt. Times New Roman font shall be used with 1.5-line spacing for typing the report.
3. 1" margin shall be left from all the sides.

4. Considering the environmental issues, students are encouraged to print on both sides of the paper.
5. The report shall be spiral bound as per the standard format of the cover page given by the Institute.
6. The report should include a Certificate (on company's letter head) from the company duly signed by the competent authority with the stamp.
7. The report shall be signed by the respective guide(s) & the Principal of the Institute 10 (Ten) days before the viva-voce examinations.
8. Student should prepare two hard bound copies of the Summer Internship Project Report and submit one copy in the institute. The other copy of the report is to be kept by the student for their record and future references.

Evaluation:

The evaluation of Internship will be done as per the following criteria:

Sr. No.	Evaluation Criteria	Marks
1	Summer Internship Report	50
2	Presentation and Viva-voce examination	50
TOTAL MARKS		100

Students must secure 36% passing marks in both components to qualify for the Diploma.

Internship and Exit:

Any student intending to exit the course must complete the summer internship (Including scoring at least 36% marks in the evaluation) before exit. A student intending to exit shall have to submit an application to Principal through Counsellor before even semester university theory examination. However, any such student desiring to withdraw the exit option may do that before the last paper of theoretical examinations of even semester.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY, CHANGA									
Smt. Chandaben Mohanbhai Patel Institute of Computer Applications									
FACULTY OF COMPUTER SCIENCE AND APPLICATIONS									
BCA, B Sc(IT), MCA, M Sc (IT) Programmes (ODD Semester)									
Academic Calendar (2024-2025)									
Week No	Month	M	T	W	T	F	S	S	Activity
1	June	24	25	26	27	28	29	30	Commencement of New Academic Sessions of BCA-III,V, B.Sc.IT-III,V, MCA-III,M.Sc.IT-III Bridge Course for BCA-I, B.Sc.IT-I (Online Mode)
2	July-August	1	2	3	4	5	6	7	Teaching Bridge Course for MCA-I, M.Sc.IT-I (Online Mode) Commencement Celebration for BCA-I, B.Sc.IT-I (Offline Mode)
3		8	9	10	11	12	13	14	Teaching Bridge Course for MCA-I, M.Sc.IT-I(Online Mode) Commencement of Acedemic sessions for BCA-I, B.Sc.IT-I
4		15	16	17	18	19	20	21	Teaching Commencement Celebration for MCA-I, M.Sc.IT-I(Offline Mode) Muharram
5		22	23	24	25	26	27	28	Teaching Commencement of Acedemic sessions for MCA-I, M.Sc.IT-I Expert Session
6		29	30	31	1	2	3	4	Teaching Expert Session
7	August-September	5	6	7	8	9	10	11	Teaching Workshop on Current Trends Aarohan Club Activity
8		12	13	14	15	16	17	18	Teaching Independence Day Vastuvit Club Activity
9		19	20	21	22	23	24	25	Teaching (Online Mode) Rakshabandhan Samarpan Club Activity
10		26	27	28	29	30	31	1	Teaching Janmashtami Samvardhan Club Activity
11	September-October	2	3	4	5	6	7	8	Teaching Industrial Visit Samvatsari(Ganesh Chaturthi)
12		9	10	11	12	13	14	15	Teaching Expert Session Tarkmanthan Club Activity
13		16	17	18	19	20	21	22	Teaching Expert Session Vignatma Club Activity
14		23	24	25	26	27	28	29	Teaching Aavishkar Club Activity
15		30	1	2	3	4	5	6	Teaching Gandhi Jyanti
16	October-November	7	8	9	10	11	12	13	Teaching Workshop on Current Trends
17		14	15	16	17	18	19	20	Teaching Sessional Examination (BCA-III,V, B.Sc.IT-III,V,MCA-III,M.Sc.IT-III)
18		21	22	23	24	25	26	27	DIWALI VACATION
19		28	29	30	31	1	2	3	
20	November - December	4	5	6	7	8	9	10	Teaching (Online Mode) University Practical Examination (BCA-III,V, B.Sc.IT-III,V, MCA-III,M.Sc.IT- III)
21		11	12	13	14	15	16	17	Teaching MCA-I, M.Sc.IT-I University Practical Examination (BCA-III,V, B.Sc.IT-III,V, MCA-III,M.Sc.IT-III) (Continue) Sessional Examination(BCA-I, B.Sc.IT-I) Guru Nanak Jayanti
22		18	19	20	21	22	23	24	Teaching MCA-I, M.Sc.IT-I University Theory Examination (BCA-III,V, B.Sc.IT-III,V, MCA-III,M.Sc.IT-III) Sessional Examination(BCA-I, B.Sc.IT-I)
23		25	26	27	28	29	30	1	University Theory Examination (BCA-III,V,B.Sc.IT-III,V, M.Sc.IT-III, MCA-III) (Continue) Universtity Practical Examination(BCA-I,B.Sc.IT-I)
24		2	3	4	5	6	7	8	Sessional Examination(MCA-I, M.Sc.IT-I) University Theory Examination (BCA-I, B.Sc.IT-I)

25	December	9	10	11	12	13	14	15	Universtity Practical Examination(MCA-I,M.Sc.IT-I) University Theory Examination (BCA-I, B.Sc.IT-I) (Continue) icSoftComp2024 International Conference Remedial Examination of BCA-III,V, B.Sc.IT-III,V, MCA-III,M.Sc.IT-III
26		16	17	18	19	20	21	22	Universtity Theory Examination(MCA-I,M.Sc.IT-I) Teaching Commencement of New Academic Sessions of BCA-IV,VI, B.Sc.IT-IV,VI, MCA-IV,M.Sc.IT-IV Remedial Examination for BCA-III,V, B.Sc.IT-III,B.Sc.IT-V, MCA-III and M.Sc.IT-III (Continue) AGNITIO
27		23	24	25	26	27	28	29	Universtity Theory Examination(MCA-I,M.Sc.IT-I) Christsmas Te
28	December-January	30	31	1	2	3	4	5	Commencement of New Academic Sessions of MCA-II,M.Sc.IT-II Teaching Remedial Examination for BCA-I, B.Sc.IT-I (Continue)

SECTION - 2

VARIOUS ADMINISTRATIVE PROCESS

Payment of tuition fees or other charges

Step : 1

- Visit University web-portal click on Pay Fees :
<https://www.charusat.ac.in/student-corner> or visit
<https://charusat.edu.in:912/FeesPaymentApp/>

Step : 2

- Enter your Student ID and Pay your Fees

Step : 3

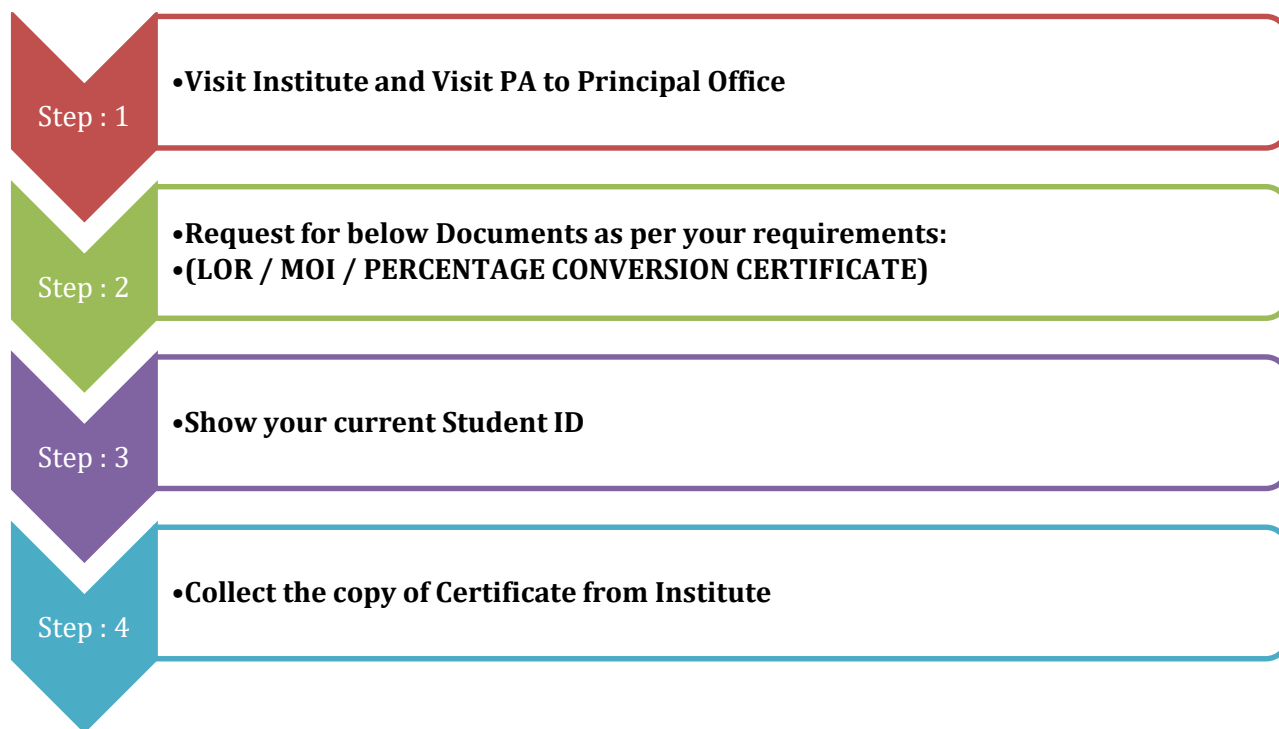
- After Successfull payment download your fees receipt

Process of Acquiring WIFI Access

To access the Wi-Fi on the Charotar University of Science and Technology (CHARUSAT) campus, students can use the following:

- **Wi-Fi enabled campus:** The entire CHARUSAT campus is Wi-Fi enabled.
- **Computers with internet:** All computers on campus have internet access.
- **Dedicated bandwidth:** Students are given a large bandwidth for internet surfing.
- **Ring connectivity:** The campus has ring connectivity to provide uninterruptable network access.

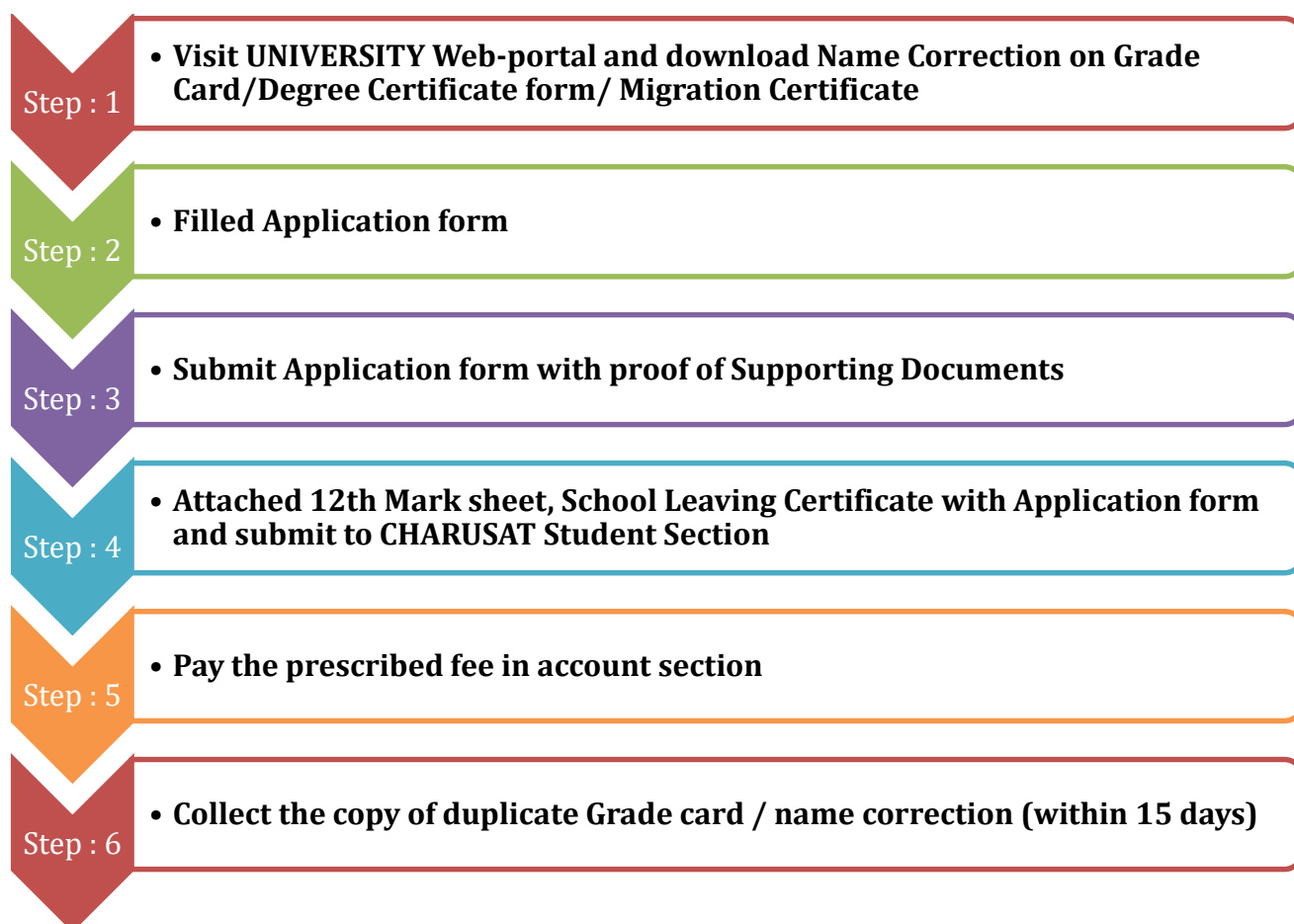
Process to obtain required Certificate from the institute



Process to obtain required Certificate from the University

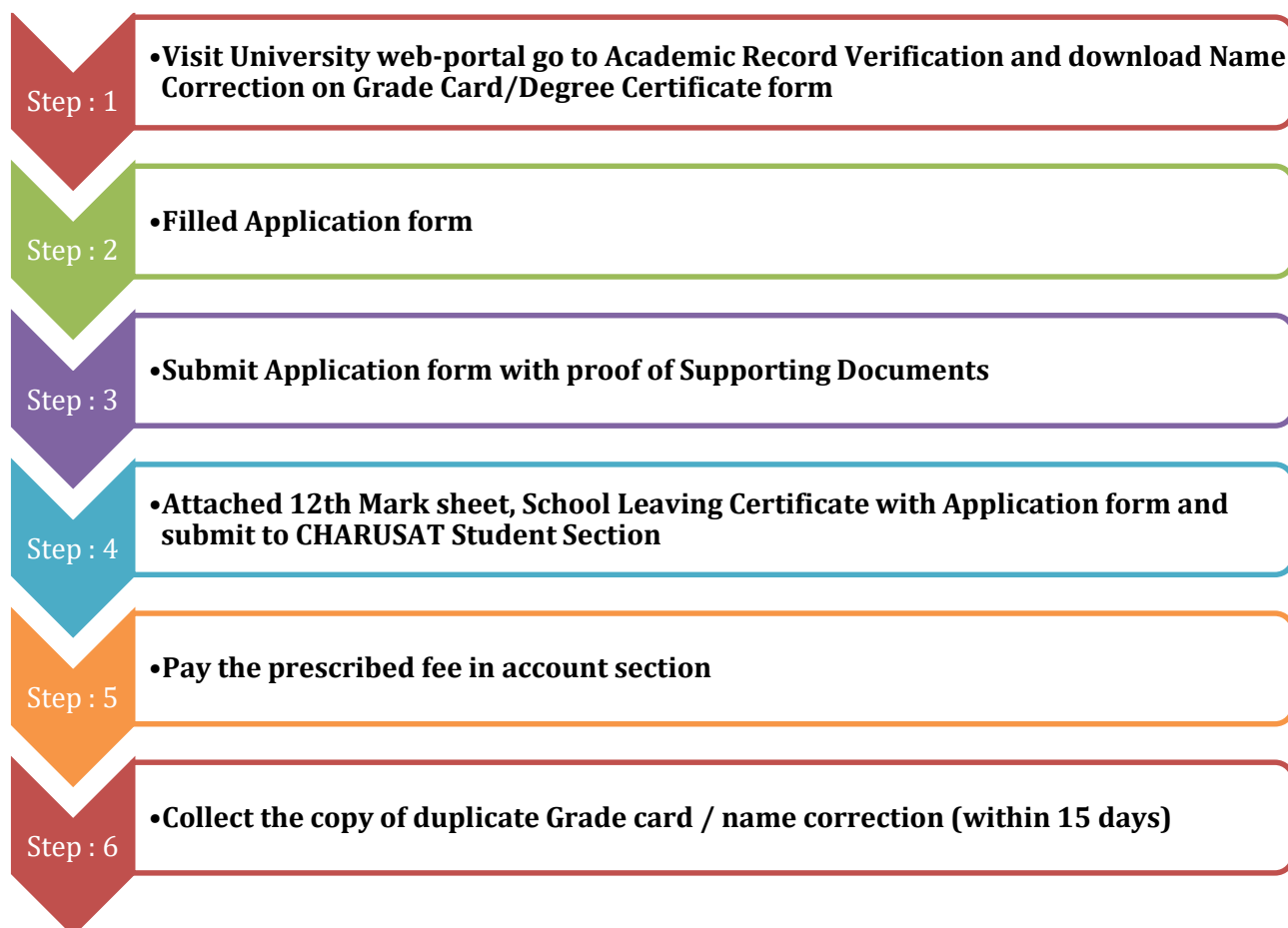
In order to obtain the required certificate at the University Level, students need to visit the Student Corner of the CHARUSAT University website. They can choose to apply online or offline and should ensure to include all necessary enclosures with their application. The application must be submitted to the Student Section of the University and should be duly attested by the relevant institute's HoD/ Principal/ Dean. Once all the necessary procedures are completed, the certificate will be issued at the University office.

Process to obtain Duplicate Grade Card/ Name correction in Grade Card/ Migration certificate

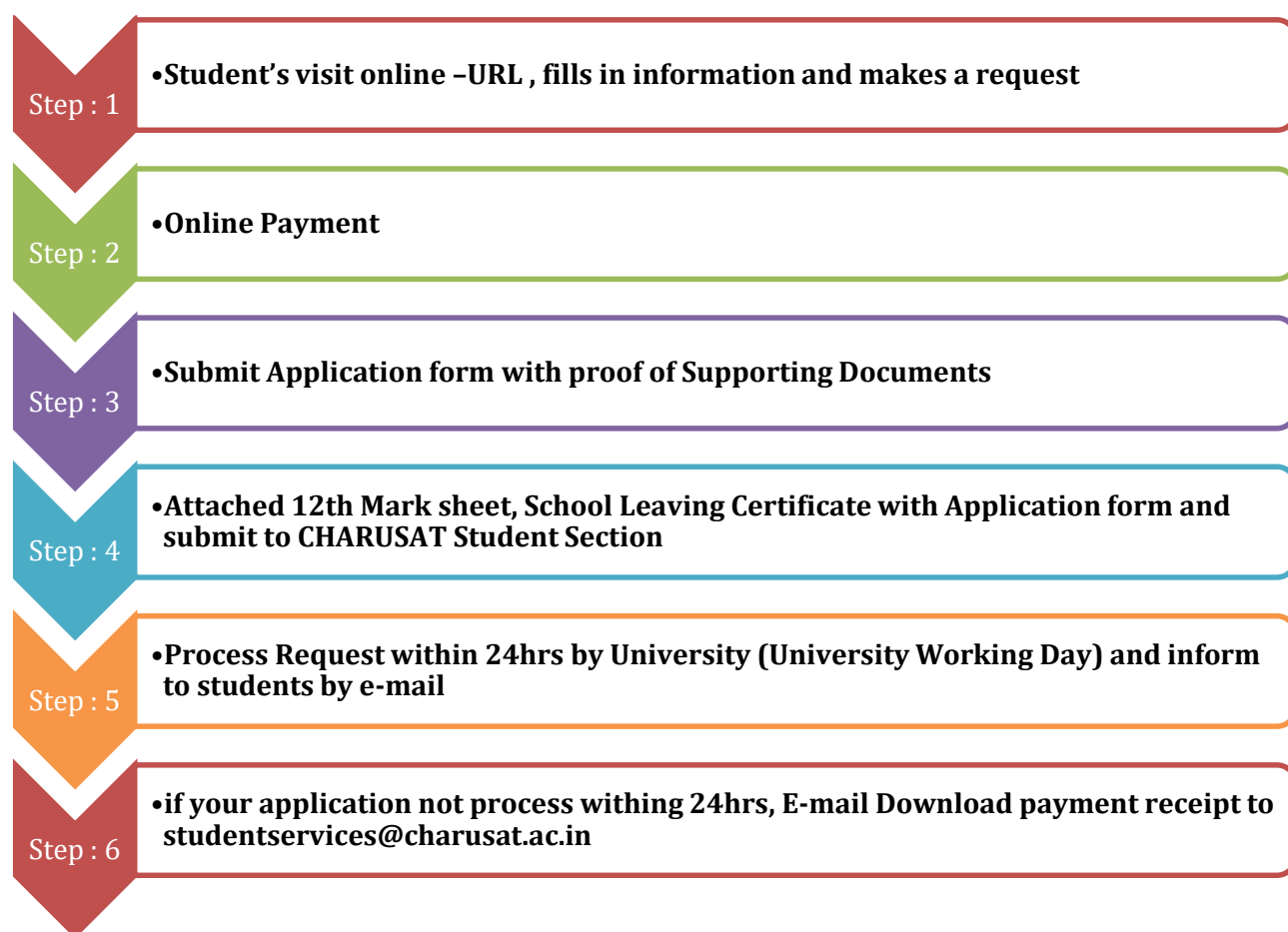


Initiator	Task
Student	Online Application Request through CHARUSAT Web Portal https://www.charusat.ac.in/student-corner Select Transcript / Duplicate Grade Card / Migration Certificate Select Document Type Migration Certificate Transcript/E-Transcript / WES Duplicate Grade Card Enter CHARUSAT Student ID Make Online Payment
Student	Pay fees at online Download Payment Receipt (for further Communication)
University Admin Office	University will get request after successful Payment (Time is depend on clearing of payment)
Entire process time duration: 24hrs after payment done	

Process for Academic Document Verification by External Agency



Work flow



SECTION – 3

UNDERTAKINGS AND DECLARATIONS

UNDERTAKING
(Observing Rules and Regulations of the University)

Roll No. _____

I, _____ Mr. /Ms. son/daughter of
_____ have secured admission at the Smt.
Chandaben Mohanbhai Patel Institute of Computer Applications of CHARUSAT
University in the academic year for the ____ Programme. We hereby confirm that we
have gone through the academic rules and regulations of the Institute very carefully and
we assure you that we will abide by the same.

Student Signature : _____

Name of the Parent/Guardian : 1. _____
2. _____

Signature of the Parent/Guardian : 1. _____ 2. _____

DECLARATION
(Code of Conducts and Disciplinary Rules)

I bearing roll no. admitted in
..... (programme) of the Institute of....., CHARUSAT
University, Changa do hereby declare and undertake that I will abide by the Code of Conduct,
including rules for misconduct/indiscipline by the students, provisions like dress code on the
campus, rules for maintaining vehicles on the campus, public display of affection (PDA),
etiquette on the campus etc.

I will abide by all the rules and regulations as and when intimated by the university and if I
am found violating any rules then, I shall be subjected to the major/minor penalties as may
deemed fit by the university.

Signature : _____

Name of the Parent/Guardian : 1. _____

2. _____

Signature of the Parent/Guardian : 1. _____ 2. _____

UNDERTAKING

(Observing Rules & Regulations of the Examination)

I, Roll No..... studying in the First year of programme at Institute of, CHARUSAT University, Changa do hereby undertake that I have read and understood all the Rules & Regulations related to Academic Dishonesty at examinations/tests/assignments and punishment in case of using unfair means, I have also gone through the Academic Regulations related to Granting of Term and Cancellation of admission, and I shall observe, follow and abide by all these rules and regulations.

I shall abide by all the rules and regulations and if I am found violating any rules then, I shall be subjected to the necessary action/penalties as per provision of rules/regulations of the university.

Signature : _____

Name : _____

Address : _____

Signature of the Parent/Guardian : 1. _____ 2. _____

UNDERTAKING
(To Refrain from Consumption of Drugs and Alcohol)

I, _____ bearing Roll No. _____ admitted in _____ (programme) at Institute of _____ do hereby declare and undertake that I will refrain myself from possession / consumption of Drugs and Alcohol.

I know that the use/possession of narcotics drugs and alcohol is a punishable offence under the law of the Government of Gujarat and if I am found guilty of using such thing/s, then it will amount to a criminal offence and I am liable for the appropriate penalty as per laws and also liable to cancel my admission from the university.

I hereby give an undertaking to the Institute that I will refrain myself from possession or consumption of Drugs and Alcohol in and around the campus.

Date :

Place :

Signature of Student

I undertake that I will take utmost care to see that my ward does not get involved in any such incident.

Name of the Parent/Guardian : 1. _

2. _

Signature of the Parent/Guardian : 1. _____ 2. _____

Address of Parent/ Guardian :

Contact no of Parent/ Guardian : 1. _____ 2. _____

4. IMPORTANT CONTACTS

+91-02697-265011 (Last 4 digits: Extension number)

Name and Designation	Extension number & Email-id
Dr. Sanskruti Patel Dean - Faculty of Computer Application	5243 Dean.fca@charusat.ac.in
Dr. Dharmendra Patel Principal	5241 Principal.cmpica@charusat.ac.in
Mr. Harikrushna Patel P.A to Principal	5242 Haripatel.mba@charusat.ac.in
Mr. Suresh patel Librarian	5255 sureshsolanki.mca@charusat.ac.in
Mr. Harikrushna Patel Student Section	5242 Haripatel.mba@charusat.ac.in
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Shri Ritesh Bhatt WIN Cell Coordinator	5106 riteshbhatt.win@charusat.ac.in
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Dr. Mrunali Patel Chairperson, Internal Complaint Committee	5163 chairperson.icc@charusat.ac.in
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